

Ramification After Blowup Is Tame

(reorganized comparison draft)

July 14, 2026

About this draft. This is a reorganized copy of the chapter for side-by-side comparison with the current version in `main.pdf`. The mathematics is unchanged; only the order of the material and the framing (dictionary tables and shared Step A–D signposts making the Kummer / Artin–Schreier / Artin–Schreier–Witt parallelism explicit) differ. Cross-references such as [Corollary 27](#) point into the other chapters and carry the numbering of the compiled thesis.

Chapter 1

Ramification After Blowup Is Tame

Throughout this chapter, let C be a smooth, projective, geometrically connected curve over k . Fix a modulus

$$\mathfrak{m} = \sum_i n_i P_i$$

on C , and set $U = C \setminus \mathfrak{m}$.

For each single-point submodulus $\mathfrak{n} = n_i P_i$ of $\mathfrak{m}$, set $d_{\mathfrak{n}} = \deg(\mathfrak{n})$. The effective divisor $\mathfrak{n}$ determines a k -rational closed point

$$Z_{\mathfrak{n}} \hookrightarrow C^{(d_{\mathfrak{n}})} = \mathrm{Sym}_k^{d_{\mathfrak{n}}}(C).$$

Let

$$\pi_{\mathfrak{n}}: X_{\mathfrak{n}} := \mathrm{Bl}_{Z_{\mathfrak{n}}}(C^{(d_{\mathfrak{n}})}) \longrightarrow C^{(d_{\mathfrak{n}})}$$

be the blowup at this point. We denote its exceptional divisor by

$$E_{\mathfrak{n}} := Z_{\mathfrak{n}} \times_{C^{(d_{\mathfrak{n}})}} X_{\mathfrak{n}}$$

By [Propositions 22](#) and [31](#), $E_{\mathfrak{n}} \cong \mathbb{P}_k^{d_{\mathfrak{n}}-1}$; in particular, it is irreducible and has codimension 1 in $X_{\mathfrak{n}}$. We denote its unique generic point by $\eta_{\mathfrak{n}}$, so that $\overline{\{\eta_{\mathfrak{n}}\}} = E_{\mathfrak{n}}$. Thus we have the Cartesian diagram

$$\begin{array}{ccc} E_{\mathfrak{n}} = \overline{\{\eta_{\mathfrak{n}}\}} & \hookrightarrow & X_{\mathfrak{n}} \\ \downarrow & & \downarrow \pi_{\mathfrak{n}} \\ Z_{\mathfrak{n}} & \hookrightarrow & C^{(d_{\mathfrak{n}})}. \end{array} \tag{1.1}$$

The purpose of this chapter is to prove [Theorem 3](#). Let $\mathcal{F}$ be the rank-one Λ -local system appearing in that theorem, and let

$$\mathcal{P} = \mathrm{Fr}(\mathcal{F})$$

be its $G = \Lambda^\times$ -torsor of frames. By [Proposition 9](#) and [definition 26](#), there is a canonical identification

$$\mathrm{Fr}(\mathcal{F}^{[d_n]}) \cong \mathcal{P}^{[d_n]}.$$

In particular, the two objects have the same ramification. It therefore suffices to prove the following torsor formulation.

Theorem 1. *Let G be a finite abelian group, and let $\mathcal{P} \rightarrow U$ be a G -torsor whose ramification is bounded by $\mathfrak{m}$. For every single-point submodule $\mathfrak{n} = n_i P_i$ of $\mathfrak{m}$, the symmetric-power torsor $\mathcal{P}^{[d_n]}$ is tamely ramified at $\eta_{\mathfrak{n}}$.*

Organization of this chapter. The chapter is arranged to make one point visible: the three cases $G = \mathbb{Z}/e\mathbb{Z}$ ($p \nmid e$), $G = \mathbb{Z}/p\mathbb{Z}$ and $G = \mathbb{Z}/p^r\mathbb{Z}$ are one and the same computation, run through Kummer, Artin–Schreier, and Artin–Schreier–Witt theory respectively.

- The *prelude* collects the background each case needs, in the order in which it is used: first the ramification theory of discrete valuation rings; then the ramification of G -torsors (the definitions that phrase everything below); and finally the three cyclic-extension theories, laid out in parallel behind a dictionary table ([Table 1.1](#)). None of it is specific to the present theorem.
- The remaining sections carry out the calculation. We first reduce [Theorem 1](#) to an explicit torsor on $\mathbb{G}_m \subset \mathbb{P}_k^1$; we then run the symmetric-power computation in all three cases at once, following the shared four-step skeleton (Steps A–D) recorded in [Table 1.2](#); and we assemble the general finite abelian case by decomposing G into its primary cyclic factors. A final section records the behaviour of ramification under products of blowups, used in [Chapter 3](#).

1.1 Prelude: Local Ramification and Cyclic Extensions

1.1.1 Ramification of discrete valuation rings

This section reviews the ramification of discrete valuations, beginning with the general theory of *discrete valuation rings* and their integral closures in finite separable extensions. We then specialize to complete rings in Galois extensions to describe the ramification filtration via lower and upper numbering, following the treatments in [\[Stacks, Tag 0EXQ\]](#) and [\[Ser79\]](#).

Definition 2 ([\[Stacks, Tag 09E4\]](#)). We say that $A \rightarrow B$ or $A \subset B$ is an extension of discrete valuation rings if A and B are discrete valuation rings and $A \rightarrow B$ is injective and local. In particular, if π_A and π_B are uniformizers of A and B , then $\pi_A = u\pi_B^e$ for some $e \geq 1$ and unit u of B . The integer e does not depend on the choice of the uniformizers as it is also the unique integer ≥ 1 such that

$$\mathfrak{m}_A B = \mathfrak{m}_B^e$$

The integer e is called the *ramification index* of B over A . We say that B is *weakly unramified* over A if $e = 1$. If the extension of residue fields $\kappa_A = A/\mathfrak{m}_A \subset \kappa_B = B/\mathfrak{m}_B$ is finite, then we set $f = [\kappa_B : \kappa_A]$ and we call it the *residual degree* or residue degree of the extension $A \subset B$.

Note that we do not require the extension of fraction fields to be finite.

Now let A be a discrete valuation ring with fraction field K , let L/K be a finite separable field extension and let $B \subset L$ be the integral closure of A in L . Then the ring extension $A \subset B$ is finite, hence B is Noetherian. The dimension of B is 1, hence B is a Dedekind domain. Let $\mathfrak{m}_1, \dots, \mathfrak{m}_n$ be the maximal ideals of B (i.e., the primes lying over $\mathfrak{m}_A$). We obtain extensions of discrete valuation rings

$$A \subset B_{\mathfrak{m}_i}$$

and hence ramification indices e_i and residue degrees f_i . We have

$$[L : K] = \sum_{i=1, \dots, n} e_i f_i$$

Note that if A is henselian (e.g. A is complete) then $n = 1$.

Definition 3 ([Stacks, Tag 09E9]). Let A be a discrete valuation ring with fraction field K . Let L/K be a finite separable extension. With B and $\mathfrak{m}_i$, $i = 1, \dots, n$ as above, we say the extension L/K is

1. unramified with respect to A if $e_i = 1$ and the extension $\kappa(\mathfrak{m}_i)/\kappa_A$ is separable for all i ,
2. tamely ramified with respect to A if either the characteristic of κ_A is 0 or the characteristic of κ_A is $p > 0$, the field extensions $\kappa(\mathfrak{m}_i)/\kappa_A$ are separable, and the ramification indices e_i are prime to p , and
3. totally ramified with respect to A if $n = 1$ and the residue field extension $\kappa(\mathfrak{m}_1)/\kappa_A$ is trivial.

If the discrete valuation ring A is clear from context, then we sometimes say L/K is unramified, totally ramified, or tamely ramified for short.

Now and for the rest of the section assume A is a *complete discrete valuation ring* with uniformizer π and residue field κ , which we assume to be perfect. When A and κ are of the same characteristic $p > 0$, then A contains a coefficient field $k \cong \kappa$ and a well known structure theorem holds: $A = k[[\pi]] \cong k[[t]]$. Let K be the fraction field of A , then $K = k((\pi))$.

For the remainder of this section, assume A is a *complete discrete valuation ring* with uniformizer π and a perfect residue field κ . In the equal characteristic case, where $\text{char}(A) = \text{char}(\kappa) = p > 0$, the Cohen Structure Theorem implies that A contains a coefficient field $k \cong \kappa$, such that $A \cong k[[\pi]] \cong k[[t]]$. Consequently, the fraction field is given by $K = k((\pi))$. Let L/K be a finite Galois extension with Galois group G . The integral closure B of A in L is itself a complete discrete valuation ring. Since the residue field κ is perfect, the extension of residue fields is separable, and there exists a uniformizer $\Pi \in B$ such that $B = A[\Pi]$.

The *lower numbering ramification filtration* of G , denoted by $(G_i)_{i \geq -1}$, is defined as:

$$G_i = \{\sigma \in G \mid v_B(\sigma(x) - x) \geq i + 1 \text{ for all } x \in B\}$$

where v_B is the valuation on L associated with B . In this indexing, $G_{-1} = G$ and G_0 corresponds to the *inertia group* of the extension L/K . Note that L/K is unramified if and only if $G_0 = \{1\}$, and it is tamely ramified if and only if $G_1 = \{1\}$.

A standard result in the theory of local fields (complete local fields with a discrete valuation and perfect residue field) ensures that the condition in the definition of G_i need only be checked for the uniformizer Π of B . Specifically, if we define the function

$$i_K^L(\sigma) = v_B(\sigma(\Pi) - \Pi)$$

for $\sigma \in G$, then $G_i = \{\sigma \in G \mid i_K^L(\sigma) \geq i + 1\}$. The subgroups G_i are normal in G and become trivial for sufficiently large i .

Consider a tower of fields $K \subset E \subset L$, and let $H = \text{Gal}(L/E) \subset G$. The lower numbering is compatible with subgroups:

$$G_i \cap H = H_i \quad \text{for all } i \geq -1$$

which follows from the identity $i_E^L = i_K^L|_H$. However, the lower numbering does not behave as simply with respect to quotients. If H is normal in G , the image of G_i in G/H is generally not $(G/H)_i$. Instead, we have $G_i H/H = (G/H)_j$, where the index j is given by the formula:

$$j = \frac{1}{e_{L/E}} \sum_{\tau \in H} \min(i_K^L(\tau), i + 1) - 1$$

In the literature, one reindexes the ramification groups by defining the Herbrand function $\phi_{L/K} : [-1, \infty) \rightarrow [-1, \infty)$:

$$\phi_{L/K}(i) = \frac{1}{e_{L/K}} \sum_{\sigma \in G} \min(i_K^L(\sigma), i + 1) - 1 = \int_0^i \frac{1}{[G_0 : G_t]} dt$$

This function is continuous, strictly increasing, and piecewise linear, making it a bijection. It satisfies the composition law $\phi_{L/K} = \phi_{E/K} \circ \phi_{L/E}$ for a tower of extensions $K \subset E \subset L$. Using this function, we define the *upper numbering ramification groups* as:

$$G^i = G_{\phi_{L/K}^{-1}(i)}$$

The primary advantage of this reindexing is its compatibility with quotients: for any normal subgroup $H \subset G$, we have:

$$G^i H/H = (G/H)^i$$

for all $i \geq -1$.

Proposition 4 ([Ser79], IV Corollary 3). *the quotients G_i/G_{i+1} , $i \geq 1$, are abelian groups, and are direct products of cyclic groups of order p . The group G_1 is a p -group.*

Proposition 5 ([Ser79], IV Corollary 4). *The inertia group G_0 has the following property: It is the semi-direct product of a cyclic group of order prime to p with a normal subgroup whose order is a power of p .*

1.1.2 Ramification of G -torsors

Generally speaking, assume we are given a locally Noetherian scheme X , a dense open $U \subset X$, a finite étale morphism $f : Y \rightarrow U$ and a prime divisor $Z \subset X$ such that $Z \cap U = \emptyset$ and the local

ring $\mathcal{O}_{X,\xi}$ of X at the generic point ξ of Z is a discrete valuation ring. Setting $K_\xi = \text{Frac}(\mathcal{O}_{X,\xi})$ we obtain a cartesian square

$$\begin{array}{ccc} \text{Spec}(K_\xi) & \longrightarrow & U \\ \downarrow & & \downarrow \\ \text{Spec}(\mathcal{O}_{X,\xi}) & \longrightarrow & X \end{array}$$

of schemes. In particular, we see that $Y \times_U \text{Spec}(K_\xi)$ is the spectrum of a finite separable algebra L_ξ/K_ξ .

Definition 6. We say:

1. Y is *unramified* over X at (Z, ξ) resp. Y is *tamely ramified* over X at (Z, ξ) if L_ξ/K_ξ is unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$ (Definition 3)
2. Y is unramified over X in codimension 1, resp. Y is tamely ramified over X in codimension 1 if L_ξ/K_ξ is unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$ for every (Z, ξ) as above.

More precisely, we decompose L_ξ into a product of finite separable field extensions of K_ξ and we require each of these to be unramified, resp. tamely ramified with respect to $\mathcal{O}_{X,\xi}$.

Now back to our original assumptions, C is projective smooth geometrically connected curve over a field k and $U = C \setminus \mathfrak{m}$. Every (Z, ξ) is now a closed point P and $\mathcal{O}_{C,P}$ is a discrete valuation ring. Assume $f : Y \rightarrow U$ is actually G -torsor $\mathcal{P} \rightarrow U$ in $U_{\text{ét}}$ for G finite abelian group.

By passing to the completion $\widehat{\mathcal{O}}_{C,P}$, we obtain the complete local field $\widehat{K}_P$. Restricting the torsor $\mathcal{P}$ to the punctured formal neighborhood $\text{Spec}(\widehat{K}_P)$ allows us to study the ramification in a strictly local context. This restriction yields a G -torsor over a field, which necessarily decomposes into a disjoint union of spectra of finite separable field extensions

$$\mathcal{P}|_{\text{Spec}(\widehat{K}_P)} \cong \bigsqcup \text{Spec}(F)$$

such that:

1. These extensions $F/\widehat{K}_P$ are all isomorphic and Galois.
2. The Galois group $H = \text{Gal}(F/\widehat{K}_P)$ identifies with the stabilizer of any connected component of the pullback $\mathcal{P}|_{\text{Spec}(\widehat{K}_P)}$, which is a subgroup of G .
3. The number of such components is then given by the index $[G : H]$.

Definition 7. We say that the extension $F/\widehat{K}_P$ has *ramification bounded by r* if the ramification group H^r is trivial. We say $\mathcal{P}$ has *ramification bounded by r at P* if the corresponding local field extensions $F/\widehat{K}_P$ satisfy this condition.

Definition 8. Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C . We say the G -torsor $\mathcal{P} \rightarrow U$ has **ramification bounded by $\mathfrak{m}$** if, for each point P_i , the local restriction $\mathcal{P}|_{\text{Spec}(\widehat{K}_{P_i})}$ has ramification bounded by the coefficient n_i .

This local property remains invariant under base change to the separable closure. Let $\bar{s} = \text{Spec}(\bar{k}) \rightarrow \text{Spec}(k)$ be a geometric point. The higher ramification groups for $\mathcal{P}|_{\text{Spec}(\widehat{K}_P)}$ are isomorphic to those of the geometric restriction $\mathcal{P}_{\bar{k}}$ at any point $\bar{x}$ lying over P . Thus an equivalent definition:

Definition 9. Let $\mathfrak{m} = \sum n_i P_i$ be a modulus on C . The G -torsor $\mathcal{P} \rightarrow U$ has **ramification bounded by $\mathfrak{m}$** if for every geometric point $\bar{x}$ of $\mathfrak{m}$ (with image $\bar{s}$ in $\text{Spec}(k)$), the restriction of $\mathcal{P}$ to

$$\text{Spec}(\widehat{\mathcal{O}_{C_{\bar{k}, \bar{x}}}}) \times_{C_{\bar{k}}} U_{\bar{k}}$$

has ramification bounded by the multiplicity of $\mathfrak{m}_{\bar{s}}$ at $\bar{x}$ in the sense of [Definition 7](#).

These definitions are equivalent because $\widehat{\mathcal{O}_{C_{\bar{k}, \bar{x}}}} \cong \widehat{\mathcal{O}_{C, P}} \otimes_k \bar{k}$. Furthermore, the notions of tame ramification and unramifiedness derived here coincide with the codimension-1 criteria in [Definition 6](#).

There is another equivariant formulation through the language of characters. The group G , being a finite abelian group over k , is étale. Consequently, the correspondence between G -torsors and the fundamental group allows us to describe G -torsors via characters. Specifically, there is an isomorphism: ([Corollary 14](#))

$$\text{Hom}_{\text{cont.}}(\pi_1^{\text{ét}}(U, \bar{x}), G) \xrightarrow{\cong} \text{Tors}(U_{\text{ét}}, G)$$

In our local setting, where we consider the restriction to the complete local field $\widehat{K}_P$, the fundamental group identifies with the absolute Galois group $G_{\widehat{K}_P} := \text{Gal}(\widehat{K}_P^{\text{sep}}/\widehat{K}_P)$. Thus, the restricted G -torsor $\mathcal{P}|_{\text{Spec}(\widehat{K}_P)}$ corresponds to a unique continuous homomorphism:

$$\rho : G_{\widehat{K}_P} \rightarrow G$$

Under this correspondence, the torsor $\mathcal{P}$ has ramification bounded by r at P if and only if the homomorphism ρ trivializes the r -th ramification group in the upper numbering:

$$\rho(G_{\widehat{K}_P}^r) = \{1\}$$

Basic Properties of Ramification of G -Torsors

We now prove some elementary lemmas regarding the ramification of G -torsors.

Lemma 10. *Let G be a finite abelian group and X be a locally Noetherian scheme over a field k . Let $U \subset X$ be a dense open subset and let (Z, ξ) be a prime divisor in the complement $X \setminus U$. Assume $\mathcal{P}_1$ and $\mathcal{P}_2$ are two G -torsors on $U_{\text{ét}}$, such that $\mathcal{P}_1$ has ramification bounded by r_1 at (Z, ξ) , and $\mathcal{P}_2$ has ramification bounded by r_2 at (Z, ξ) . Then their product $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ has ramification bounded by $\max(r_1, r_2)$ at (Z, ξ) .*

Proof. Let $A = \widehat{\mathcal{O}_{X, \xi}}$ and let $K = \text{Frac}(A)$. Let $\rho_1, \rho_2 : G_K \rightarrow G$ be the associated continuous homomorphisms corresponding to the G -torsors $\mathcal{P}_1|_{\text{Spec}(K)}, \mathcal{P}_2|_{\text{Spec}(K)}$. We finish by noticing that the associated character of $(\mathcal{P}_1 \otimes^G \mathcal{P}_2)|_{\text{Spec}(K)}$ is $\rho = \rho_1 + \rho_2$. $\square$

Lemma 11. *Let $f : X \rightarrow Y$ be a morphism of locally Noetherian schemes. Let $U_X \subset X$ and $U_Y \subset Y$ be dense open subschemes such that $f^{-1}(U_Y) \subset U_X$.*

Let (Z_X, ξ_X) and (Z_Y, ξ_Y) be prime divisors of X and Y such that $Z_X \cap U_X = \emptyset$ and $Z_Y \cap U_Y = \emptyset$. Suppose $f(\xi_X) = \xi_Y$ and that f is étale at ξ_X . Let $\mathcal{P}$ be a G -torsor over U_Y , and let $f^{-1}\mathcal{P}$ be its pullback to $f^{-1}(U_Y) \subset U_X$. Then, the ramification of $f^{-1}\mathcal{P}$ at ξ_X is bounded by r if and only if the ramification of $\mathcal{P}$ at ξ_Y is bounded by r .

Proof. The boundedness of ramification is determined by the behavior of the torsor over the completion of the local rings at the generic points. Let $A = \mathcal{O}_{Y,\xi}$ and $B = \mathcal{O}_{X,\xi_X}$ be the discrete valuation rings at the generic points, with fraction fields K and L respectively. Since f is étale at ξ_X , the map $A \rightarrow B$ is a flat, unramified (hence weakly unramified) local homomorphism. Consequently, the extension of completions $\widehat{L}/\widehat{K}$ is a finite unramified extension of complete discretely valued fields. We finish by noticing that the upper numbering filtration on the absolute Galois group is compatible with unramified base change.¹ $\square$

Let C be a smooth, proper (projective) curve over a field k , and let $U \subset C$ be a dense open subset. Let G be a finite abelian group. and let $\pi_U : U' \rightarrow U$ be a G -torsor in $U_{\text{ét}}$. Let C' be the normal proper model of U' over k (i.e. the unique smooth proper curve which has U' as a dense open subset). Let $\pi : C' \rightarrow C$ be the induced morphism. (which is finite generically étale over U). Let $U'^{(d)}$ be the standard d -th symmetric product of U' as a scheme, and let $U'^{[d]}$ be the torsor-theoretic symmetric product, by [Corollary 27](#) we have a canonical morphism $q_U : U'^{(d)} \rightarrow U'^{[d]}$. Note that $C'^{(d)}$ is smooth and finite over $C^{(d)}$, hence it is the normalization of $C^{(d)}$ inside $K(C'^{(d)})$. $U'^{(d)}$ is a dense open subset of $C'^{(d)}$, hence $C'^{(d)}$ is a normal proper model of $U'^{(d)}$. Let $C'^{[d]}$ be the normalization of $C^{(d)}$ in the function field $K(U'^{[d]})$, then q_U extends to a map $q : C'^{(d)} \rightarrow C'^{[d]}$, and we have the following diagrams:

$$\begin{array}{ccc} U' & \hookrightarrow & C' \\ \pi_U \downarrow & & \downarrow \pi \\ U & \hookrightarrow & C \end{array} \qquad \begin{array}{ccc} U'^{(d)} & \hookrightarrow & C'^{(d)} \\ q_U \downarrow & & \downarrow q \\ U'^{[d]} & \hookrightarrow & C'^{[d]} \\ \downarrow & & \downarrow \\ U^{(d)} & \hookrightarrow & C^{(d)} \end{array}$$

Lemma 12. *In the context of the preceding discussion, the morphism $U'^{(d)} \rightarrow U'^{[d]} \hookrightarrow C'^{[d]}$ is unramified in codimension 1 (see [Definition 6](#)).*

Proof. Let $C' \setminus U' = \{x_1, \dots, x_m\}$. For each $x \in C' \setminus U'$, define the locus

$$Z_x = \{D \in C'^{(d)} \mid x \in \text{Supp}(D)\}.$$

Each Z_x is a prime divisor in the symmetric product $C'^{(d)}$, and the boundary $C'^{(d)} \setminus U'^{(d)}$ is precisely the finite union $\bigcup_{i=1}^m Z_{x_i}$.

Let η be the generic point of Z_x . It suffices to show that the local ring extension $\mathcal{O}_{C'^{[d]}, q(\eta)} \hookrightarrow \mathcal{O}_{C'^{(d)}, \eta}$ is weakly unramified ([Definition 2](#)).

Let $p : C'^d \rightarrow C'^{(d)}$ be the natural quotient. Consider the closed subvariety

$$H = \{x\} \times C' \times \dots \times C' \subset C'^d.$$

¹Specifically, let $G_K = \text{Gal}(K^{\text{sep}}/K)$ and $G_L = \text{Gal}(L^{\text{sep}}/L)$. For an unramified extension, the Herbrand function is the identity, which implies that for any $r \geq 0$:

$$G_L^r = G_K^r \cap G_L$$

The subvariety H is irreducible, and its image under p is exactly Z_x . Consequently, p maps the generic point η_H of H to the generic point η of Z_x .

$$\begin{array}{ccccc}
\eta_H & \longrightarrow & H & \longrightarrow & C'^d \\
\downarrow & & \downarrow & & \downarrow p \\
\eta & \longrightarrow & Z_x & \longrightarrow & C'^{(d)} \\
\downarrow & & \downarrow & & \downarrow q \\
q(\eta) & \longrightarrow & q(Z_x) & \longrightarrow & C'^{[d]}
\end{array}$$

It suffices to show that the local ring extension $\mathcal{O}_{C'^{[d]}, q(\eta)} \hookrightarrow \mathcal{O}_{C'^d, \eta_H}$ is weakly unramified. Let $R = \mathcal{O}_{C'^d, \eta_H}$. And let $A = \mathcal{O}_{C'^{[d]}, q(\eta)}$.

Recall that $C'^{[d]}$ is the quotient of C'^d by the finite group $\Xi = K_G \rtimes S_d$ (see [Corollary 27](#)), and the morphism $q \circ p : C'^d \rightarrow C'^{[d]}$ is *branched Galois Cover* (finite, surjective map of normal integral schemes $X \rightarrow Y$, with function field extension $K(X)/K(Y)$ Galois).

Hence the decomposition group of η_H in Ξ and the inertia group of η_H in Ξ are defined:

$$D_{\eta_H} = \{g \in \Xi \mid g(\eta_H) = \eta_H\}$$

$$I_{\eta_H} = \{g \in D_{\eta_H} \mid g^*(u) \equiv u \pmod{\mathfrak{m}_R} \text{ for all } u \in R\}$$

Claim: The inertia group I_{η_H} is trivial.

Let $K = K(C')$ be the function field of C' , then $K(C'^d) = K^{\otimes_k d} \cong \text{Frac}(K_1 \otimes_k K_2 \otimes_k \cdots \otimes_k K_d)$ Where K_i is the copy of K corresponding to the i -th factor of C'^d . The closed subscheme H corresponds to the prime ideal $\mathfrak{p}_H \subset \mathcal{O}_{C'^d}$, and $R = \mathcal{O}_{C'^d, \mathfrak{p}_H}$. The residue field of R is exactly $K(H) \cong \text{Frac}(k(x) \otimes_k K_2 \otimes_k \cdots \otimes_k K_d)$ where $k(x)$ is the residue field of x .

The action of $\gamma = (g_1, \dots, g_d, \sigma) \in \Xi$, via the pullback (γ^*) on $K(C'^d)$ does two things (in that order):

1. It permutes the tensor factors according to σ . A function strictly in K_i is mapped to a function in $K_{\sigma(i)}$.
2. It applies the field automorphism $g_i^* \in \text{Aut}(K/k)$ to the respective factors.

Let $\gamma \in I_{\eta_H} \subset \Xi$, Then the induced automorphism $\bar{\gamma}^*$ on $K(H)$ must be the identity map. This implies that σ must fix i for every $i \geq 2$ (The K_i are linearly independent), and hence $\sigma = id$. The trivial action of $\bar{\gamma}^*$ restricts to a trivial action of g_i^* on K_i for every $i \geq 2$, and hence $g_i = 1$ for every $i \geq 2$ (since the action of G on $K(C')$ is faithful). Finally, since $g_1 g_2 \cdots g_d = 1$, we also have $g_1 = 1$. Since the inertia group I_{η_H} is trivial, the local ring extension $A \hookrightarrow R$ is weakly unramified. (See for example [\[Stacks, Tag 09E3\]](#), [\[Stacks, Tag 0BTF\]](#))

□

1.1.3 Cyclic extensions: Kummer, Artin–Schreier, and Artin–Schreier–Witt

The three classes of cyclic extensions we use share one shape. In each, a cyclic group A sits in a short exact sequence of commutative group schemes $0 \rightarrow A \rightarrow \mathcal{G} \xrightarrow{\psi} \mathcal{G} \rightarrow 0$ for the étale topology

(a Lang-type isogeny), so that ψ -torsors over a field K are classified by $\mathcal{G}(K)/\psi\mathcal{G}(K)$ via additive Hilbert 90; and the ramification of the associated extension of a local field is read off from the pole order(s) of a normalized defining datum. The following dictionary records the correspondence; the three columns are quoted in detail below.

	Kummer	Artin–Schreier	Artin–Schreier–Witt
group A	$\mathbb{Z}/e\mathbb{Z}, p \nmid e$	$\mathbb{Z}/p\mathbb{Z}$	$\mathbb{Z}/p^r\mathbb{Z}$
on K	contains μ_e	$\text{char } K = p$	$\text{char } K = p$
isogeny ψ	$t \mapsto t^e$ on $\mathbb{G}_m$	$\wp = t^p - t$ on $\mathbb{G}_a$	$\wp = F - \text{id}$ on $\mathbb{W}_r$
sequence	$1 \rightarrow \mu_e \rightarrow \mathbb{G}_m \xrightarrow{e} \mathbb{G}_m \rightarrow 1$	$0 \rightarrow \mathbb{Z}/p\mathbb{Z} \rightarrow \mathbb{G}_a \xrightarrow{\wp} \mathbb{G}_a \rightarrow 0$	$0 \rightarrow \mathbb{Z}/p^r\mathbb{Z} \rightarrow \mathbb{W}_r \xrightarrow{\wp} \mathbb{W}_r \rightarrow 0$
classifies	$K^\times/(K^\times)^e$	$K/\wp K$	$W_r(K)/\wp W_r(K)$
break from	$v_K(a) \bmod e$	pole order $m = -v_K(a)$	Schmid–Witt $\max_j p^{r-1-j}m_j$
reference	Theorem 13	Theorem 14	Theorem 24 , Theorem 28

Table 1.1: The three cyclic-extension theories in parallel.

Kummer and Artin–Schreier theory

For cyclic extensions of degree prime to p (Kummer) and of degree p (Artin–Schreier), the following two theorems make the “break from” row of [Table 1.1](#) precise: the valuation of a defining element $a \in K$ determines the ramification index and the jumps in the upper numbering filtration. Throughout this section, let K be a discrete valuation field with a perfect residue field κ of characteristic $p > 0$.

Theorem 13 (Ramification in Kummer Extensions, [[Koc97](#)], Proposition 1.83). *Assume K contains the n -th roots of unity μ_n . Let L/K be the extension given by the equation $X^n = a$ for some $a \in K^\times$ and denote by G its Galois group. Then we have:*

1. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in the residue field κ is an n -th power, the extension L/K is trivial.*
2. *If $v_K(a) \in n\mathbb{Z}$ and the image of $a\pi^{-v_K(a)}$ in the residue field κ is not an n -th power, the extension L/K is cyclic and unramified.*
3. *If $v_K(a) \notin n\mathbb{Z}$, the extension L/K is cyclic and ramified. Specifically, if $\gcd(|v_K(a)|, n) = 1$, the extension is totally ramified of degree n . Otherwise it has ramification index $\frac{n}{\gcd(v_K(a), n)}$.*

Conversly, Kummer theory ensures that every cyclic extension of degree n , prime to p of a field that contains n -th roots of unity, is of the above form. Moreover, in the above we can always take $a \in \mathcal{O}_K$

Note that in the case of total ramification the extension is tamely ramified.

Theorem 14 (Ramification in Artin-Schreier Extensions, [[Tho05](#)]). *Let $\wp(x) = x^p - x$ be the Artin-Schreier operator. Let L/K be the extension given by the equation $X^p - X = a$ for some $a \in K$ and denote by G its Galois group. Then we have:*

1. *If $v_K(a) > 0$ or if $v_K(a) = 0$ and $a \in \wp(K)$, the extension L/K is trivial.*

2. If $v_K(a) = 0$ and if $a \notin \wp(K)$, the extension L/K is cyclic of degree p and unramified.
3. If $v_K(a) = -m < 0$ with $m \in \mathbb{Z}_{>0}$ and if m is prime to p , the extension L/K is cyclic of degree p again and totally ramified. Moreover, its ramification groups are given by:

$$G = G^{(-1)} = \dots = G^{(m)} \quad \text{and} \quad G^{(m+1)} = 1.$$

Conversely, Artin-Schreier theory ensures that every cyclic extension of degree p takes this form. Moreover, in the above and under the isomorphism $K \cong k((t))$, one can always take a of the form $ct^{-m} + a_{-m+1}t^{-m+1} + \dots + a_{-1}t^{-1} + a_0$. If k is algebraically closed then there is a change of variables such that $a = u^{-m}$.

Artin–Schreier–Witt theory

This is the third column of [Table 1.1](#): it generalizes Artin–Schreier theory from $\mathbb{Z}/p\mathbb{Z}$ to $\mathbb{Z}/p^r\mathbb{Z}$ by replacing the field K with the ring $W_r(K)$ of length- r Witt vectors, the operator $\wp(u) = u^p - u$ with $\wp = F - \text{id}$, and the pole-order break criterion of [Theorem 14](#) with the Schmid–Witt formula ([Theorem 28](#)). The development is correspondingly longer, since the ring W_r itself must be set up; a reader willing to take [Theorem 24](#) and [Theorem 28](#) on faith may skip to [Theorem 32](#).

Artin–Schreier theory, together with the ramification formula of [Theorem 14](#), gives complete control over the cyclic extensions of degree p of a local field of characteristic p . To treat the groups $G = \mathbb{Z}/p^r\mathbb{Z}$ we need the analogous control over cyclic extensions of degree p^r , which is provided by Witt’s generalization: the field K is replaced by the ring $W_r(K)$ of p -typical Witt vectors of length r , the operator $\wp(u) = u^p - u$ by its Witt-vector analogue $\wp = F - \text{id}$, and the classical ramification-break formula by the Schmid–Witt formula. This subsection collects the definitions and the elementary lemmas that we use; our references for the general theory are [\[Ser79, II, §6\]](#) and [\[Rab14\]](#).

Throughout, p is a fixed prime, and $r \geq 1$ is an integer. All the Witt vectors appearing in this thesis are p -typical and of finite length.

The ring of Witt vectors

For $n \geq 0$, the n -th **Witt polynomial** in the variables $X_0, \dots, X_n$ is

$$w_n(X_0, \dots, X_n) = \sum_{j=0}^n p^j X_j^{p^{n-j}} = X_0^{p^n} + pX_1^{p^{n-1}} + \dots + p^n X_n.$$

Theorem 15 (Witt; [\[Ser79, II, §6, Theorem 6\]](#), [\[Rab14, §1\]](#)). *There exists a unique family of polynomials $S_n, P_n \in \mathbb{Z}[X_0, \dots, X_n; Y_0, \dots, Y_n]$, $n \geq 0$, such that*

$$w_n(S_0, \dots, S_n) = w_n(X_0, \dots, X_n) + w_n(Y_0, \dots, Y_n), \quad w_n(P_0, \dots, P_n) = w_n(X_0, \dots, X_n) \cdot w_n(Y_0, \dots, Y_n)$$

for all n . For any commutative ring A , the operations

$$\vec{a} \oplus \vec{b} := (S_n(\vec{a}; \vec{b}))_{0 \leq n \leq r-1}, \quad \vec{a} \odot \vec{b} := (P_n(\vec{a}; \vec{b}))_{0 \leq n \leq r-1}$$

make the set A^r into a commutative ring $W_r(A)$, the ring of **p -typical Witt vectors of length r over A** , with zero $(0, \dots, 0)$ and unit $(1, 0, \dots, 0)$. Additive inverses are likewise given by universal integral polynomials; we write $\ominus$ for Witt subtraction. The **ghost map**

$$w: W_r(A) \longrightarrow A^r, \quad \vec{a} \longmapsto (w_0(\vec{a}), w_1(\vec{a}), \dots, w_{r-1}(\vec{a}))$$

is a ring homomorphism to the product ring A^r ; it is injective when p is a non-zero-divisor in A , and bijective when $p \in A^\times$.

We index the components of a Witt vector $\vec{a} = (a_0, \dots, a_{r-1})$ by $0, \dots, r-1$. For $r = 1$ the polynomials reduce to $S_0 = X_0 + Y_0$ and $P_0 = X_0 Y_0$, so $W_1(A) = A$ as a ring.

Since the ring operations are given by universal polynomials with integer coefficients, W_r is a functor: a ring homomorphism $\sigma: A \rightarrow B$ induces the ring homomorphism

$$W_r(\sigma): W_r(A) \rightarrow W_r(B), \quad W_r(\sigma)(a_0, \dots, a_{r-1}) = (\sigma a_0, \dots, \sigma a_{r-1}),$$

acting *componentwise*. Two consequences of this trivial-looking observation are used repeatedly and deserve emphasis: a ring automorphism σ of A commutes with $\oplus$ and $\ominus$ on $W_r(A)$, and a Witt vector $\vec{a} \in W_r(A)$ satisfies $W_r(\sigma)(\vec{a}) = \vec{a}$ if and only if *each component* a_n is σ -invariant. In particular, for a group Γ acting on A by ring automorphisms,

$$W_r(A)^\Gamma = W_r(A^\Gamma) \quad \text{inside } W_r(A). \quad (1.2)$$

The uniqueness statement in [Theorem 15](#) is an instance of a general principle that we invoke several times, so we record it explicitly.

Lemma 16 (ghost principle). *Let Φ and Ψ be two operations on Witt vectors given by universal polynomials with integer coefficients in the components of their arguments. If $w \circ \Phi = w \circ \Psi$ holds over the polynomial ring $\mathbb{Z}[X_0, X_1, \dots]$ on the components of the arguments, then $\Phi = \Psi$ over every commutative ring.*

Proof. Over $\mathbb{Z}[X_0, X_1, \dots]$ the prime p is a non-zero-divisor, so the ghost map is injective by [Theorem 15](#) and $\Phi = \Psi$ there; this is an identity of integral polynomials, which persists under every base change $\mathbb{Z}[X_0, X_1, \dots] \rightarrow A$. $\square$

The only structural feature of the addition polynomials S_n that we shall need is their “triangular” shape:

Proposition 17 (shape of Witt addition). *For every $n \geq 0$ there is a polynomial $C_n \in \mathbb{Z}[X_0, \dots, X_{n-1}; Y_0, \dots, Y_{n-1}]$ involving only the variables of index strictly less than n , such that*

$$(\vec{a} \oplus \vec{b})_n = a_n + b_n + C_n(a_0, \dots, a_{n-1}; b_0, \dots, b_{n-1}). \quad (1.3)$$

We refer to C_n as the **carry polynomial**. The analogous statement holds for $\ominus$.

Proof. By induction on n , using the recursion defining S_n :

$$p^n S_n = \sum_{j \leq n} p^j (X_j^{p^{n-j}} + Y_j^{p^{n-j}}) - \sum_{j < n} p^j S_j^{p^{n-j}}.$$

The terms with $j = n$ contribute $p^n(X_n + Y_n)$, and all remaining terms involve only variables and polynomials S_j of index $j < n$, which by induction lie in $\mathbb{Z}[X_{<n}; Y_{<n}]$. Since S_n has integer coefficients ([Theorem 15](#)), $C_n := S_n - X_n - Y_n$ is an integral polynomial in the variables of index $< n$. The statement for $\ominus$ follows by the same argument applied to the inversion polynomials. $\square$

Frobenius, Verschiebung and truncation

From now on, A denotes an $\mathbb{F}_p$ -algebra; this is the only case we use. Three operators act on the rings $W_r(A)$:

Definition 18. Let A be an $\mathbb{F}_p$ -algebra.

1. The **Frobenius** $F: W_r(A) \rightarrow W_r(A)$ is $F := W_r(\varphi_A)$, where $\varphi_A: a \mapsto a^p$ is the absolute Frobenius of A . Explicitly, F acts componentwise: $F(a_0, \dots, a_{r-1}) = (a_0^p, \dots, a_{r-1}^p)$. By functoriality, F is a ring endomorphism of $W_r(A)$.
2. The **Verschiebung** is the shift

$$V: W_{r-1}(A) \rightarrow W_r(A), \quad V(a_0, \dots, a_{r-2}) = (0, a_0, \dots, a_{r-2}).$$

Iterating, $V^j: W_{r-j}(A) \rightarrow W_r(A)$; in particular $V^{r-1}: A = W_1(A) \rightarrow W_r(A)$ sends $c \mapsto (0, \dots, 0, c)$.

3. The **truncation** is

$$t: W_r(A) \rightarrow W_{r-1}(A), \quad t(a_0, \dots, a_{r-1}) = (a_0, \dots, a_{r-2}).$$

Remark. For a general (not necessarily $\mathbb{F}_p$ -) algebra A , the Frobenius is defined by universal polynomials characterized by $w_n \circ F = w_{n+1}$, and the componentwise formula above holds precisely in characteristic p ; see [Ser79, II, §6] or [Rab14, §2]. We will not need the general case.

We record from the general theory ([Ser79, II, §6], [Rab14, §2]) how multiplication by p interacts with these operators.

Proposition 19. Let A be an $\mathbb{F}_p$ -algebra. Then:

1. t is a natural ring homomorphism commuting with F , and its kernel is

$$\ker(t: W_r(A) \rightarrow W_{r-1}(A)) = V^{r-1}A = \{(0, \dots, 0, c) : c \in A\}.$$

2. V is additive, and $FV = VF$.
3. Multiplication by p on $W_r(A)$ equals $V \circ F = F \circ V$ (composed with the appropriate truncation); explicitly,

$$p \cdot (a_0, \dots, a_{r-1}) = (0, a_0^p, \dots, a_{r-2}^p).$$

Iterating, $p^j \cdot \vec{a} = (0, \dots, 0, a_0^{p^j}, \dots, a_{r-1-j}^{p^j})$; in particular $p^r = 0$ on $W_r(A)$, and

$$p^{r-1} \cdot (a_0, \dots, a_{r-1}) = (0, \dots, 0, a_0^{p^{r-1}}). \quad (1.4)$$

Proof. (1) That t is a ring homomorphism follows from the recursive definition of the S_n, P_n : the first $r-1$ components of a sum or product depend only on the first $r-1$ components of the arguments (Proposition 17). Naturality and the commutation with the (componentwise) F are clear, and the kernel is computed componentwise. (2) Additivity of V follows from Lemma 16 and the ghost identity $w_n(V\vec{a}) = p w_{n-1}(\vec{a})$ (with $w_{-1} := 0$); the commutation $FV = VF$ is immediate since F is componentwise. (3) The identity $FV = p$ holds over arbitrary rings for the general Frobenius, by Lemma 16 applied to $w_n(FV\vec{a}) = w_{n+1}(V\vec{a}) = p w_n(\vec{a}) = w_n(p\vec{a})$; in characteristic p the general Frobenius is componentwise, giving the displayed formula. See [Ser79, II, §6, Proposition 8 ff.] for details. $\square$

Example 20. $W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$ canonically. Indeed, by (1.4) the unit $\underline{1} = (1, 0, \dots, 0)$ satisfies $p^j \cdot \underline{1} = (0, \dots, 0, 1, 0, \dots, 0)$ (1 in position j), so $\underline{1}$ has additive order p^r ; since $|W_r(\mathbb{F}_p)| = p^r$, the additive group is cyclic of order p^r generated by the unit, and $m \mapsto m \cdot \underline{1}$ is a ring isomorphism $\mathbb{Z}/p^r\mathbb{Z} \xrightarrow{\cong} W_r(\mathbb{F}_p)$. We write $\underline{m} \in W_r(\mathbb{F}_p)$ for the image of $m \in \mathbb{Z}/p^r\mathbb{Z}$. (Similarly, the full ring of Witt vectors satisfies $W(\mathbb{F}_p) \cong \mathbb{Z}_p$; we shall not need this.)

Definition 21. The **Artin–Schreier–Witt operator** on $W_r(A)$, A an $\mathbb{F}_p$ -algebra, is

$$\wp := F - \text{id}, \quad \wp \vec{u} = F\vec{u} \ominus \vec{u}.$$

For $r = 1$ this is the classical operator $\wp(u) = u^p - u$ of Theorem 14.

Since F is a ring endomorphism, $\wp$ is an endomorphism of the additive group $(W_r(A), \oplus)$: $\wp(\vec{u} \oplus \vec{v}) = \wp \vec{u} \oplus \wp \vec{v}$. Moreover $\wp$ commutes with $W_r(\sigma)$ for every ring homomorphism σ (both F and the ring operations are natural), and with V and t by Proposition 19.

The following bookkeeping lemma is elementary but does the essential work of reducing statements about W_r to statements about W_{r-1} and about A itself: the truncation t and the inclusion V^{r-1} of its kernel slice W_r into lower-length pieces, compatibly with $\wp$, and *addition of an element of the kernel is carry-free*.

Lemma 22 (Witt bookkeeping). *Let A be an $\mathbb{F}_p$ -algebra.*

- (i) $t: W_r(A) \rightarrow W_{r-1}(A)$ is a ring homomorphism commuting with F , hence with $\wp$; its kernel is $V^{r-1}A$.
- (ii) $V^{r-1}: A \rightarrow W_r(A)$ is additive, and $\wp \circ V^{r-1} = V^{r-1} \circ \wp$, where on the left $\wp$ is the operator on $W_r(A)$ and on the right the classical $\wp(c) = c^p - c$ on A .
- (iii) (carry-free top-level addition) For all $\vec{a} \in W_r(A)$ and $c \in A$,

$$\vec{a} \oplus V^{r-1}(c) = (a_0, \dots, a_{r-2}, a_{r-1} + c).$$

Proof. (i) and (ii) restate parts of Proposition 19 (for (ii): V^{r-1} is additive and commutes with the componentwise F , hence with $\wp = F - \text{id}$). (iii) By Lemma 16 it suffices to verify the identity over $\mathbb{Z}[a_0, \dots, a_{r-1}, c]$, where the ghost map is injective. Put $\vec{b} := (0, \dots, 0, c)$ and $\vec{s} := \vec{a} \oplus \vec{b}$. For $n < r - 1$ we have $w_n(\vec{b}) = 0$, so $w_n(\vec{s}) = w_n(\vec{a})$, and by induction on n (solving for s_n , using that p is a non-zero-divisor) $s_n = a_n$ for $n \leq r - 2$. For $n = r - 1$, $w_{r-1}(\vec{b}) = p^{r-1}c$ gives $p^{r-1}s_{r-1} = p^{r-1}a_{r-1} + p^{r-1}c$, i.e. $s_{r-1} = a_{r-1} + c$. $\square$

Remark. Lemma 22(iii) fails for the addition of vectors supported in lower positions: in general the carry polynomials C_n of (1.3) are non-zero, and Witt addition is *not* componentwise. In particular, operations performed componentwise on a Witt vector — such as discarding the regular parts of Laurent-series components — do *not* in general respect Witt addition, and will be carefully avoided.

The Artin–Schreier–Witt isogeny and its torsors

The componentwise ring operations make $\mathbb{A}_{\mathbb{F}_p}^r$ a ring scheme, the **Witt vector group scheme** $\mathbb{W}_r$: its underlying scheme is $\mathbb{A}_{\mathbb{F}_p}^r = \text{Spec } \mathbb{F}_p[T_0, \dots, T_{r-1}]$, with the group law given by the addition polynomials S_n , so that $\mathbb{W}_r(A) = W_r(A)$ functorially. The operator $\wp = F - \text{id}$ is an endomorphism

of the group scheme $\mathbb{W}_r$, and it is the Witt-vector instance of a *Lang isogeny*: by [Example 20](#) and the lemma below, it sits in a short exact sequence of group schemes over $\mathbb{F}_p$ for the étale topology

$$0 \longrightarrow \mathbb{Z}/p^r\mathbb{Z} \cong W_r(\mathbb{F}_p) \longrightarrow \mathbb{W}_r \xrightarrow{\wp} \mathbb{W}_r \longrightarrow 0. \quad (1.5)$$

The corresponding covers admit the following completely explicit description, which is the form in which we use them. For $\vec{g} \in W_r(A)$ we write $A[\vec{X}]/(\wp\vec{X} \ominus \vec{g})$, $\vec{X} = (X_0, \dots, X_{r-1})$, for the quotient of $A[X_0, \dots, X_{r-1}]$ by the ideal generated by the r components of the Witt vector $\wp\vec{X} \ominus \vec{g} \in W_r(A[X_0, \dots, X_{r-1}])$.

Lemma 23 (ASW covers are étale torsors). *Let A be an $\mathbb{F}_p$ -algebra and $\vec{g} \in W_r(A)$. Let*

$$B := A[\vec{X}]/(\wp\vec{X} \ominus \vec{g}).$$

Then B is finite free over A with basis $\{\prod_n X_n^{a_n} : 0 \leq a_n \leq p-1\}$, and étale over A . The translation action of $W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$,

$$m: \vec{X} \longmapsto \vec{X} \oplus \underline{m},$$

makes $\text{Spec } B$ a $\mathbb{Z}/p^r\mathbb{Z}$ -torsor over $\text{Spec } A$.

Proof. By [\(1.3\)](#) and the componentwise formula for F , the n -th defining equation has the triangular form

$$X_n^p - X_n - g_n + C'_n(X_0, \dots, X_{n-1}; g_0, \dots, g_{n-1}) = 0$$

for universal integral polynomials C'_n . Triangularity gives the free basis; the Jacobian matrix of the system is lower triangular with diagonal entries $pX_n^{p-1} - 1 = -1$, a unit, so B is étale over A . Finally, $\text{Spec } B$ is the pullback along $\vec{g}: \text{Spec } A \rightarrow \mathbb{W}_r$ of the covering $\wp: \mathbb{W}_r \rightarrow \mathbb{W}_r$; by the triangular form just established (over $A = \mathbb{F}_p[T_0, \dots, T_{r-1}]$, $\vec{g} = (T_0, \dots, T_{r-1})$), $\wp$ is a surjective étale homomorphism of group schemes, and its kernel is $\{\vec{u} : F\vec{u} = \vec{u}\} = W_r(\mathbb{F}_p)$ since F is componentwise. A surjective étale group homomorphism is a torsor under its kernel, and torsors pull back to torsors; hence $\text{Spec } B$ is a $W_r(\mathbb{F}_p)$ -torsor over $\text{Spec } A$. $\square$

Artin–Schreier–Witt theory for fields

Over a field, the sequence [\(1.5\)](#) computes all cyclic extensions of p -power degree, exactly as classical Artin–Schreier theory does for degree p .

Theorem 24 (Artin–Schreier–Witt theory; [\[Tho05, §3–4\]](#)). *Let $M \supseteq \mathbb{F}_p$ be a field, M^s a separable closure, and $\Gamma = \text{Gal}(M^s/M)$. Then $\wp$ is surjective on $W_r(M^s)$ with kernel $W_r(\mathbb{F}_p)$, and there is a canonical isomorphism of abelian groups*

$$\delta: W_r(M)/\wp W_r(M) \xrightarrow{\cong} \text{Hom}_{\text{cont}}(\Gamma, \mathbb{Z}/p^r\mathbb{Z}) = H^1(M, \mathbb{Z}/p^r\mathbb{Z}), \quad [\vec{g}] \longmapsto (\sigma \mapsto \sigma(\vec{u}) \ominus \vec{u}),$$

where $\vec{u} \in W_r(M^s)$ is any solution of $\wp\vec{u} = \vec{g}$. Under the correspondence between characters and torsors ([Corollary 14](#)), the class $[\vec{g}]$ corresponds to the $\mathbb{Z}/p^r\mathbb{Z}$ -torsor

$$\text{Spec } M[\vec{X}]/(\wp\vec{X} \ominus \vec{g})$$

of [Lemma 23](#). Moreover, for a class of order p^s in $W_r(M)/\wp W_r(M)$, the fixed field $M(\vec{u}) := M(u_0, \dots, u_{r-1})$ of $\ker \delta[\vec{g}]$ is a cyclic extension of M of degree p^s ; the torsor is connected if and only if $[\vec{g}]$ has order p^r , in which case $\text{Spec } M[\vec{X}]/(\wp\vec{X} \ominus \vec{g}) \cong \text{Spec } M(\vec{u})$ is (the spectrum of) a cyclic field extension of degree p^r . Every cyclic extension of M of degree dividing p^r arises this way.

Sketch of proof. Surjectivity of $\wp$ on $W_r(M^s)$ and the computation of its kernel follow from the triangular equations in the proof of [Lemma 23](#): each component of a preimage is obtained by solving a separable equation $X_n^p - X_n = (\dots)$, and $F\vec{u} = \vec{u}$ holds if and only if $u_n^p = u_n$ for all n , i.e. $\vec{u} \in W_r(\mathbb{F}_p)$. Given $\vec{g}$ and a solution $\wp\vec{u} = \vec{g}$, for $\sigma \in \Gamma$ we have $\wp(\sigma\vec{u} \ominus \vec{u}) = \sigma\vec{g} \ominus \vec{g} = 0$ (using (1.2) and that $W_r(\sigma)$ commutes with $\wp$), so $\sigma\vec{u} \ominus \vec{u} \in W_r(\mathbb{F}_p)$; one checks it is a continuous homomorphism in σ (the kernel is fixed pointwise by Γ), independent of the choices modulo nothing and of the representative $\vec{g}$ modulo $\wp W_r(M)$. Injectivity: if $\delta[\vec{g}] = 0$ then $\vec{u}$ is Γ -invariant, hence lies in $W_r(M)$ by (1.2), so $\vec{g} = \wp\vec{u} \in \wp W_r(M)$. Surjectivity of δ is the additive Hilbert 90: $H^1(\Gamma, W_r(M^s)) = 0$, by induction on r along the exact sequence $0 \rightarrow M^s \xrightarrow{V^{r-1}} W_r(M^s) \xrightarrow{t} W_{r-1}(M^s) \rightarrow 0$ of Γ -modules ([Lemma 22\(i\)](#)) and the classical case $H^1(\Gamma, M^s) = 0$. The remaining statements follow from [Theorem 13](#) applied to the character $\delta[\vec{g}]$, whose image has the same order as the class $[\vec{g}]$. For details see [[Tho05](#), §3–4] or [[Rab14](#)]. $\square$

The order of a class is partially visible in its 0-th component:

Corollary 25. *Let $M \supseteq \mathbb{F}_p$ be a field and $\vec{g} \in W_r(M)$ with $g_0 = 0$. Then $p^{r-1}[\vec{g}] = 0$ in $W_r(M)/\wp W_r(M)$; equivalently, the cyclic extension attached to $[\vec{g}]$ has degree at most p^{r-1} . Consequently, if $[\vec{g}]$ has order p^r (e.g. if $\vec{g}$ defines a connected torsor), then every representative of the class has non-zero 0-th component.*

Proof. By (1.4), $p^{r-1}\vec{g} = (0, \dots, 0, g_0^{p^{r-1}}) = 0$ already in $W_r(M)$. $\square$

Remark. When $M = k((x))$ with k algebraically closed, every finite separable extension L/M is totally ramified: the residue field extension is trivial (as k is algebraically closed) and $\mathcal{O}_L$ is a complete discrete valuation ring, so $[L : M] = ef = e$. Thus a class of order p^r in $W_r(k((x)))/\wp$ corresponds to a cyclic, *totally ramified* extension of degree p^r .

Isobaricity of the Witt addition polynomials

The following homogeneity property of the universal addition polynomials is what makes degree estimates on Witt sums possible: even though the carry polynomials C_n of (1.3) are complicated, they are *isobaric* for the weighting in which the j -th component has weight p^j .

Lemma 26 (isobaricity of Witt addition). *Assign to the variables X_j and Y_j the weight p^j . Then the n -th addition polynomial S_n is isobaric of weight p^n : every monomial of S_n has total weight exactly p^n . The same holds for each component of a d -fold iterated Witt sum $\bigoplus_{i=1}^d \vec{a}_i$, viewed as a universal polynomial in the components $a_{i,j}$ of weight p^j .*

Proof. The n -th ghost polynomial $w_n = \sum_{j \leq n} p^j X_j^{p^{n-j}}$ is isobaric of weight p^n for this weighting. The recursion

$$p^n S_n = \sum_{j \leq n} p^j (X_j^{p^{n-j}} + Y_j^{p^{n-j}}) - \sum_{j < n} p^j S_j^{p^{n-j}}$$

preserves isobaricity of weight p^n by induction on n : each S_j is isobaric of weight p^j , so $S_j^{p^{n-j}}$ is isobaric of weight p^n . Isobaricity persists under iterated substitution of Witt sums into Witt sums (substituting an isobaric polynomial of weight p^j for a variable of weight p^j preserves weights), giving the statement for d -fold sums. $\square$

Ramification breaks: the Schmid–Witt formula

Finally we state the generalization of the ramification formula of [Theorem 14](#) to Artin–Schreier–Witt extensions. As in the classical case, the formula reads off the largest upper-numbering ramification break from the pole orders of a defining Witt vector — provided the representative is normalized so that no interference between the levels can occur. The relevant normalization is the following.

Definition 27. Let $K = k((x))$. For $g \in K$ put $m(g) := \max(0, -v_x(g))$ (the pole order of g , or 0 if g is regular). A vector $\vec{g} = (g_0, \dots, g_{r-1}) \in W_r(K)$ is **reduced** if for every j , either $m(g_j) = 0$ or $p \nmid m(g_j)$.

Theorem 28 (Schmid–Witt break formula). *Let $K = k((x))$ with k perfect of characteristic p , and let L/K be a cyclic totally ramified extension of degree p^r whose Artin–Schreier–Witt class ([Theorem 24](#)) is represented by a reduced vector $\vec{g} \in W_r(K)$, with $m_j := m(g_j)$. Then the largest upper-numbering ramification break of L/K equals*

$$u = \max_{0 \leq j \leq r-1} p^{r-1-j} m_j.$$

This is due, for finite residue fields, to Kanesaka–Sekiguchi [[KS79](#)] and Thomas [[Tho05](#)] (building on Schmid [[Sch37](#)] for $r = 1$), and for arbitrary perfect residue fields to Elder–Keating [[EK25](#)]. We use it as stated, in one direction only: *evaluating the formula on one reduced representative computes u* ; no minimization over representatives is needed.

Remark. Reducedness is exactly what makes the formula exact. If two levels $j < j'$ with $m_j, m_{j'} \neq 0$ satisfied $p^{r-1-j}m_j = p^{r-1-j'}m_{j'}$, then $m_{j'} = p^{j'-j}m_j$ would be divisible by p , contradicting [Definition 27](#); hence the maximum is attained at a unique level, and no cancellation of breaks between levels can occur.

Remark. For $r = 1$, a reduced vector is a single element $g \in K$ whose pole order m_0 is zero or prime to p , and [Theorem 28](#) recovers [Theorem 14](#): the largest (indeed the only) upper break is m_0 . The existence of reduced representatives — and, over $K = k((x))$, of reduced representatives whose components are polynomials in x^{-1} without constant term — is not part of the general theory quoted here; it will be proved where it is used, by induction on r through the truncation maps of [Lemma 22](#).

1.2 Reduction to $\mathbb{P}_k^1$

We argue that it is enough to prove [Theorem 1](#) for $C = \mathbb{P}_k^1$, $\mathfrak{n} = d \cdot 0$, and $\mathcal{P} \rightarrow \mathbb{G}_m = \mathbb{P}_k^1 \setminus \{0, \infty\}$ a G -torsor given by an explicit global equation. Informally, this is because *the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ is a formal–local invariant of $\mathcal{P}$ at P* : it depends only on the completed local ring $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$ together with the restriction of $\mathcal{P}$ to the punctured formal disc $\mathrm{Spec}(k((x)))$, since all of the constructions involved — self products, quotients by finite groups, and blowups — commute with the flat base change to the completion; for blowups, this is [Theorem 30](#). We emphasize that this is not a general principle that can simply be quoted: it is a statement about the particular constructions at hand, and it is proved by checking each construction separately. This is the content of [Lemma 29](#) and [Corollary 30](#) below. The second ingredient of the reduction is [Lemma 31](#): every local torsor class appearing in [Theorem 1](#) is realized by an explicit torsor over $\mathbb{G}_m \subset \mathbb{P}_k^1$.

Throughout this section we assume that k is algebraically closed; in particular every closed point of C is k -rational, and $\mathfrak{n} = dP$ with $P \in C(k)$ and $d = \deg \mathfrak{n}$.

Remark. The assumption that k is algebraically closed is harmless in our situation, for the following reasons.

1. **(Base change to $\bar{k}$.)** Assume k is perfect, so that $\bar{k} = k^{sep}$. The formation of $C^{(d)}$ (a quotient by a finite group), of the blowup $X_{\mathfrak{n}}$ along $Z_{\mathfrak{n}}$, and of $\mathcal{P}^{[d]}$ (Corollary 27) commutes with the flat base change $\mathrm{Spec}(\bar{k}) \rightarrow \mathrm{Spec}(k)$. Since $E_{\mathfrak{n}} \cong \mathbb{P}_k^{\deg n - 1}$ is geometrically irreducible, $(E_{\mathfrak{n}})_{\bar{k}}$ is irreducible and its generic point $\bar{\eta}$ lies over $\eta_{\mathfrak{n}}$; moreover $\mathcal{O}_{X_{\mathfrak{n}}, \eta_{\mathfrak{n}}} \rightarrow \mathcal{O}_{(X_{\mathfrak{n}})_{\bar{k}}, \bar{\eta}}$ is a weakly unramified extension of discrete valuation rings with separable residue field extension. The proof of Lemma 11 uses only these two properties of the extension (there the extension was moreover finite étale), so it applies verbatim: the ramification of $(\mathcal{P}_{\bar{k}})^{[d]} = (\mathcal{P}^{[d]})_{\bar{k}}$ at $\bar{\eta}$ is bounded by r if and only if the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ is. Hence unramifiedness and tameness at $\eta_{\mathfrak{n}}$ may be checked after base change to $\bar{k}$.
2. **(Rationality of P — a caveat.)** If P is a closed point with $\kappa(P) \neq k$, then after base change to $\bar{k}$ the point $Z_{\mathfrak{n}}$ corresponds to the divisor $\sum_{\bar{P} \mapsto P} n \bar{P}$ on $C_{\bar{k}}$, whose support consists of $[\kappa(P) : k]$ distinct points, whereas the computation below treats a modulus supported at a single rational point. We therefore prove Theorem 1 under the standing assumption that k is algebraically closed; this suffices for our purposes, since in the proof of Theorem 76 the field is assumed algebraically closed before the present theorem is invoked.

We now formulate the formal–local invariance precisely. Fix $P \in C(k)$ lying in the support of $\mathfrak{m}$, set $\mathfrak{n} = dP$, and fix a uniformizer x of $\mathcal{O}_{C, P}$; it induces an isomorphism $\widehat{\mathcal{O}_{C, P}} \cong k[[x]]$. The scheme $\mathrm{Spec}(k[[x]])$ has two points — the closed point, mapping to P , and the generic point, mapping to the generic point of C — so the punctured formal disc $\mathrm{Spec}(k((x)))$ maps to U , and we denote by

$$\mathcal{P}_x := \mathcal{P} \times_U \mathrm{Spec}(k((x)))$$

the corresponding restriction of $\mathcal{P}$; it is a G -torsor over $\mathrm{Spec}(k((x)))$. Let $e_1, \dots, e_d \in k[[x_1, \dots, x_d]]$ be the elementary symmetric polynomials in $x_1, \dots, x_d$, and set

$$V = \mathrm{Spec}\left(k[[e_1, \dots, e_d]][e_d^{-1}]\right), \quad W = \mathrm{Spec}\left(k[[x_1, \dots, x_d]][(x_1 \cdots x_d)^{-1}]\right).$$

The inclusion $k[[e_1, \dots, e_d]] \subset k[[x_1, \dots, x_d]]$ induces a finite flat morphism $\hat{r} : W \rightarrow V$ (note that inverting $e_d = x_1 \cdots x_d$ inverts each x_i), and for each i we let

$$q_i : W \rightarrow \mathrm{Spec}(k((x))), \quad x \mapsto x_i.$$

The group S_d acts on W over V by permuting $x_1, \dots, x_d$, and $q_{\sigma(i)} = q_i \circ \sigma$.

Lemma 29 (Formal–local invariance). *In the above notation:*

1. The choice of x induces an isomorphism $\widehat{\mathcal{O}_{C^{(d)}, Z_{\mathfrak{n}}}} \cong k[[e_1, \dots, e_d]]$ carrying the maximal ideal to $(e_1, \dots, e_d)$, such that the induced morphism $c : \mathrm{Spec}(k[[e_1, \dots, e_d]]) \rightarrow C^{(d)}$ is flat and satisfies $c^{-1}(U^{(d)}) = V$; we write $c_V : V \rightarrow U^{(d)}$ for the restriction.
2. There is an induced isomorphism $\widehat{\mathcal{O}_{X_{\mathfrak{n}}, \eta_{\mathfrak{n}}}} \cong \kappa[[e_d]]$, where $\kappa = k(\frac{e_1}{e_d}, \dots, \frac{e_{d-1}}{e_d})$. In particular $\hat{K}_{\eta_{\mathfrak{n}}} := \mathrm{Frac}(\widehat{\mathcal{O}_{X_{\mathfrak{n}}, \eta_{\mathfrak{n}}}}) \cong \kappa((e_d))$, and the canonical morphism $\mathrm{Spec}(\hat{K}_{\eta_{\mathfrak{n}}}) \rightarrow U^{(d)}$ factors as $\mathrm{Spec}(\kappa((e_d))) \rightarrow V \xrightarrow{c_V} U^{(d)}$, where the first map does not depend on $(C, \mathcal{P})$.

3. There is a canonical isomorphism of G -torsors over V

$$\mathcal{P}^{[d]} \times_{U^{(d)}} V \cong \Xi \setminus \left(q_1^{-1} \mathcal{P}_x \times_W \cdots \times_W q_d^{-1} \mathcal{P}_x \right).$$

Consequently, the G -torsor $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_n})}$ — and hence whether the ramification of $\mathcal{P}^{[d]}$ at η_n is bounded by r (resp. tame, resp. unramified) — depends only on d and on the isomorphism class of the G -torsor $\mathcal{P}_x$ over $k((x))$, and not otherwise on the pair $(C, \mathcal{P})$.

Proof. (1) Since P is a k -rational point of the smooth curve C , the uniformizer x induces $\widehat{\mathcal{O}_{C,P}} \cong k[[x]]$. As completion at rational points takes products of k -varieties to completed tensor products, we get $\widehat{\mathcal{O}_{C^d, (P, \dots, P)}} \cong k[[x]] \widehat{\otimes}_k \cdots \widehat{\otimes}_k k[[x]] = k[[x_1, \dots, x_d]]$. Write $A = \mathcal{O}_{C^{(d)}, Z_n}$ and $B = \mathcal{O}_{C^d, (P, \dots, P)}$. Since k is algebraically closed, a closed point $(Q_1, \dots, Q_d) \in C^d$ maps to $Z_n = dP$ under the finite projection $r : C^d \rightarrow C^{(d)}$ if and only if $Q_1 = \cdots = Q_d = P$; hence $(P, \dots, P)$ is the unique point of C^d over Z_n , $B = (r_* \mathcal{O}_{C^d})_{Z_n}$ is a finite A -module, and $A = B^{S_d}$ (formation of invariants commutes with localization at invariant primes). Since $\mathfrak{m}_B$ is the unique maximal ideal of B over $\mathfrak{m}_A$, we have $\mathfrak{m}_B^N \subset \mathfrak{m}_A B \subset \mathfrak{m}_B$ for some N , so the $\mathfrak{m}_A$ -adic and $\mathfrak{m}_B$ -adic topologies on B agree, and therefore $B \otimes_A \hat{A} \cong \hat{B} \cong k[[x_1, \dots, x_d]]$. Now $A = B^{S_d}$ is the kernel of $B \rightarrow \prod_{\sigma \in S_d} B$, $b \mapsto (\sigma(b) - b)_\sigma$; as $\hat{A}$ is flat over A , tensoring with $\hat{A}$ preserves this kernel, so

$$\hat{A} \cong (B \otimes_A \hat{A})^{S_d} \cong k[[x_1, \dots, x_d]]^{S_d}.$$

Finally, $k[[x_1, \dots, x_d]]^{S_d} = k[[e_1, \dots, e_d]]$: the ring $k[x_1, \dots, x_d]$ is a finite free module over $k[e_1, \dots, e_d]$, the same topological argument identifies $k[[x_1, \dots, x_d]]$ with $k[x_1, \dots, x_d] \otimes_{k[e_1, \dots, e_d]} k[[e_1, \dots, e_d]]$, and taking S_d -invariants — again a kernel, preserved by the flat base change $k[e_1, \dots, e_d] \rightarrow k[[e_1, \dots, e_d]]$ — yields $k[[e_1, \dots, e_d]]$. Under these identifications the maximal ideal of $\hat{A}$ is $(e_1, \dots, e_d)$, and c is flat as the composition of the flat morphisms $\mathrm{Spec}(\hat{A}) \rightarrow \mathrm{Spec}(A) \rightarrow C^{(d)}$.

It remains to check that $c^{-1}(C^{(d)} \setminus U^{(d)}) = V(e_d)$ as sets. We have $C^{(d)} \setminus U^{(d)} = \bigcup_{Q \in \mathrm{Supp}(\mathfrak{m})} Z_Q$ with $Z_Q = \{D : Q \in \mathrm{Supp}(D)\}$, as in the proof of Lemma 12. For $Q \neq P$ the divisor Z_Q does not contain the point $Z_n = dP$, so its ideal is not contained in $\mathfrak{m}_A$, and $c^{-1}(Z_Q) = \emptyset$. For $Q = P$, pull back along the finite surjective morphism $\mathrm{Spec}(\hat{B}) \rightarrow \mathrm{Spec}(\hat{A})$: the preimage of Z_P in C^d is $\bigcup_i p_i^{-1}(P)$, whose trace on $\mathrm{Spec}(\hat{B}) = \mathrm{Spec}(k[[x_1, \dots, x_d]])$ is $V(x_1) \cup \cdots \cup V(x_d) = V(x_1 \cdots x_d) = V(e_d)$; by surjectivity, $c^{-1}(Z_P) = V(e_d)$ as claimed. Hence $c^{-1}(U^{(d)}) = V$, and likewise $\mathrm{Spec}(\hat{B}) \times_{C^d} U^d = W$.

(2) Blowing up commutes with flat base change, and the center Z_n pulls back under c to the closed point $V(e_1, \dots, e_d)$; hence

$$\hat{Y} := \mathrm{Bl}_{(e_1, \dots, e_d)}(\mathrm{Spec} k[[e_1, \dots, e_d]]) \cong X_n \times_{C^{(d)}} \mathrm{Spec}(\hat{A}),$$

and the exceptional divisor $\hat{E} \subset \hat{Y}$ is the base change of E_n along $\mathrm{Spec}(\hat{A}/\hat{\mathfrak{m}}) = \mathrm{Spec}(\kappa(Z_n))$, i.e. the projection $\mathrm{pr} : \hat{Y} \rightarrow X_n$ restricts to an isomorphism $\hat{E} \cong E_n \cong \mathbb{P}_k^{d-1}$. Let $\hat{\eta}$ be the generic point of $\hat{E}$. Then $\mathrm{pr}(\hat{\eta}) = \eta_n$, and the induced local homomorphism of discrete valuation rings $\mathcal{O}_{X_n, \eta_n} \rightarrow \mathcal{O}_{\hat{Y}, \hat{\eta}}$ carries a uniformizer to a uniformizer (the ideal of E_n pulls back to the ideal of $\hat{E}$) and induces an isomorphism of residue fields (both are the function field of $\mathbb{P}_k^{d-1}$); hence it induces an isomorphism on completions. The completion of $\mathcal{O}_{\hat{Y}, \hat{\eta}}$ is computed exactly as for the blowup of affine space at the origin, which we carry out in detail below: on the chart $u_d \neq 0$ we have $e_i = \frac{u_i}{u_d} e_d$, the local ring at $\hat{\eta}$ is $\left(\hat{A} \left[\frac{u_1}{u_d}, \dots, \frac{u_{d-1}}{u_d} \right] \right)_{(e_d)}$, its residue field is $\kappa = k\left(\frac{u_1}{u_d}, \dots, \frac{u_{d-1}}{u_d} \right) = k\left(\frac{e_1}{e_d}, \dots, \frac{e_{d-1}}{e_d} \right)$, and its completion

is $\kappa[[e_d]]$. Finally, e_d is invertible in $\widehat{K}_{\eta_n} = \kappa((e_d))$, so the composite $\mathrm{Spec}(\widehat{K}_{\eta_n}) \rightarrow \mathrm{Spec}(\mathcal{O}_{\hat{Y}, \hat{\eta}}) \rightarrow \hat{Y} \rightarrow \mathrm{Spec}(\hat{A})$ factors through V , giving the asserted factorization of $\mathrm{Spec}(\widehat{K}_{\eta_n}) \rightarrow U^{(d)}$ through c_V .

(3) Let $h : \mathcal{P}^{\times_k d} \rightarrow U^{(d)}$ denote the canonical morphism; it is finite, being the composition of the finite étale morphism $\mathcal{P}^{\times_k d} \rightarrow U^d$ with the finite morphism $U^d \rightarrow U^{(d)}$. By Corollary 27, $\mathcal{P}^{[d]} = \Xi \backslash \mathcal{P}^{\times_k d} = \underline{\mathrm{Spec}}_{U^{(d)}} \left((h_* \mathcal{O}_{\mathcal{P}^{\times_k d}})^\Xi \right)$. Pushforward along the affine morphism h commutes with the flat base change $c_V : V \rightarrow U^{(d)}$, and the formation of Ξ -invariants is a kernel, hence also commutes with flat base change; therefore

$$\mathcal{P}^{[d]} \times_{U^{(d)}} V \cong \Xi \backslash \left(\mathcal{P}^{\times_k d} \times_{U^{(d)}} V \right).$$

By step (1), $U^d \times_{U^{(d)}} V \cong W$, so $\mathcal{P}^{\times_k d} \times_{U^{(d)}} V \cong \mathcal{P}^{\times_k d} \times_{U^d} W$. For each i , the composition $W \rightarrow \mathrm{Spec}(k[[x_1, \dots, x_d]]) \rightarrow \mathrm{Spec}(k[[x]]) \rightarrow C$ — the middle map being the completion of the projection $p_i : C^d \rightarrow C$, i.e. $x \mapsto x_i$ — factors through $\mathrm{Spec}(k((x))) \rightarrow U$, since x_i is invertible on W ; this composite is precisely q_i , and it commutes with $p_i : U^d \rightarrow U$. Since $\mathcal{P}^{\times_k d} = p_1^{-1} \mathcal{P} \times_{U^d} \dots \times_{U^d} p_d^{-1} \mathcal{P}$, base changing to W gives an isomorphism

$$\mathcal{P}^{\times_k d} \times_{U^d} W \cong q_1^{-1} \mathcal{P}_x \times_W \dots \times_W q_d^{-1} \mathcal{P}_x,$$

which is equivariant for the actions of G^d (in particular of K_G) and of S_d — the latter acting on W by permuting the x_i , compatibly with $q_{\sigma(i)} = q_i \circ \sigma$. Passing to Ξ -quotients yields the displayed isomorphism.

For the final statement: the schemes V and W , the maps q_i and $\hat{r}$, the Ξ -action, and the point $\mathrm{Spec}(\kappa((e_d))) \rightarrow V$ of (2) are all constructed from d alone; by (3), the torsor $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_n})}$ is the pullback along $\mathrm{Spec}(\kappa((e_d))) \rightarrow V$ of a torsor constructed from $\mathcal{P}_x$ and these data. Since the ramification at η_n is by definition read off from $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_n})}$ together with the discrete valuation ring $\kappa[[e_d]]$ (Definition 6, Definition 7), it depends only on $(\mathcal{P}_x, d)$. $\square$

Corollary 30 (Reduction to $\mathbb{P}^1$). *Let C, C' be smooth projective geometrically connected curves over the algebraically closed field k , let $\mathcal{P} \rightarrow U = C \setminus \mathfrak{m}$ and $\mathcal{P}' \rightarrow U' = C' \setminus \mathfrak{m}'$ be G -torsors, let $P \in \mathrm{Supp}(\mathfrak{m})(k)$ and $P' \in \mathrm{Supp}(\mathfrak{m}')(k)$, and let x, x' be uniformizers at P, P' respectively. Assume that $\mathcal{P}_x \cong \mathcal{P}'_{x'}$ as G -torsors, compatibly with the isomorphism $k((x)) \cong k((x'))$, $x \mapsto x'$. Then for every $d \geq 1$ and every r , the ramification of $\mathcal{P}^{[d]}$ at η_{dP} is bounded by r if and only if the ramification of $\mathcal{P}'^{[d]}$ at $\eta_{dP'}$ is; in particular, the one is unramified (resp. tamely ramified) at η_{dP} if and only if the other is at $\eta_{dP'}$.*

Proof. Immediate from Lemma 29: both $\mathcal{P}^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{dP}})}$ and $\mathcal{P}'^{[d]}|_{\mathrm{Spec}(\widehat{K}_{\eta_{dP'}})}$ are identified, compatibly with the discrete valuation ring $\kappa[[e_d]]$, with the pullback along $\mathrm{Spec}(\kappa((e_d))) \rightarrow V$ of $\Xi \backslash (q_1^{-1} \mathcal{P}_x \times_W \dots \times_W q_d^{-1} \mathcal{P}_x)$. $\square$

Lemma 31 (Realization over $\mathbb{G}_m$). *Let k be algebraically closed of characteristic $p > 0$, and let $\mathcal{Q}$ be a G -torsor over $\mathrm{Spec}(k((x)))$.*

1. *If $\mathcal{Q}$ is tamely ramified, then there exist an integer e prime to p , an injective homomorphism $\iota : \mathbb{Z}/e\mathbb{Z} \hookrightarrow G$, and an integer a prime to e , such that*

$$\mathcal{P}' := \iota_* \left(\mathrm{Spec} \left(k[x, x^{-1}][T]/(T^e - x^a) \right) \right)$$

is a G -torsor over $\mathbb{G}_m$, tamely ramified at 0 and ∞ , with $\mathcal{P}'_x \cong \mathcal{Q}$. Here $T^e = x^a$ is regarded as a $\mathbb{Z}/e\mathbb{Z}$ -torsor via a fixed primitive e -th root of unity $\zeta_e \in k$, and $\iota_* \mathcal{Q}' := (\mathcal{Q}' \times G)/(\mathbb{Z}/e\mathbb{Z})$ denotes the induced G -torsor.

2. If $G = \mathbb{Z}/p\mathbb{Z}$ and $\mathcal{Q}$ has ramification bounded by d (Definition 7), then there exists $f \in x^{-1}k[x^{-1}]$, either $f = 0$ or $f = cx^{-m} + a_{-m+1}x^{-m+1} + \dots + a_{-1}x^{-1}$ with $c \neq 0$, $p \nmid m$ and $m < d$, such that

$$\mathcal{P}' := \text{Spec}\left(k[x, x^{-1}][X]/(X^p - X - f)\right)$$

is a $\mathbb{Z}/p\mathbb{Z}$ -torsor over $\mathbb{G}_m$ with $\mathcal{P}'_x \cong \mathcal{Q}$. Moreover $\mathcal{P}'$ extends to a torsor over $\mathbb{P}_k^1 \setminus \{0\}$ — in particular it is unramified at ∞ — and it has ramification bounded by d at 0.

In both cases $\mathcal{P}'_x$ denotes the restriction of $\mathcal{P}'$ to $\text{Spec}(k((x)))$, as before.

Proof. (1) Since k is algebraically closed, the residue field of $k((x))$ is algebraically closed, so $k((x))$ admits no non-trivial unramified extensions, and its tamely ramified extensions are exactly the Kummer extensions $k((x^{1/e}))$ with $p \nmid e$ (Theorem 13). Hence the tame quotient G^t of $\text{Gal}(k((x))^{sep}/k((x)))$ is procyclic, $G^t \cong \varprojlim_{p \nmid e} \mu_e(k) \cong \prod_{\ell \neq p} \mathbb{Z}_\ell$; fix a topological generator γ . Let $\rho : \text{Gal}(k((x))^{sep}/k((x))) \rightarrow G$ be the continuous homomorphism classifying $\mathcal{Q}$. Tame-ness means that ρ kills the wild inertia subgroup, so ρ factors through G^t and is determined by $g := \rho(\gamma)$, whose order e is prime to p . Let $\iota : \mathbb{Z}/e\mathbb{Z} \xrightarrow{\sim} \langle g \rangle \subset G$ be given by $1 \mapsto g$, so that $\rho = \iota \circ \chi$ for a surjective character $\chi : G^t \rightarrow \mathbb{Z}/e\mathbb{Z}$. On the other hand, for $a \in \mathbb{Z}$ the equation $T^e = x^a$ defines a $\mu_e \cong \mathbb{Z}/e\mathbb{Z}$ -torsor over $\text{Spec}(k((x)))$, and by Kummer theory the classes of x^a , $a \in \mathbb{Z}/e\mathbb{Z}$, exhaust $\text{Hom}_{\text{cont.}}(G^t, \mathbb{Z}/e\mathbb{Z}) \cong \mathbb{Z}/e\mathbb{Z}$, the class of x^a being surjective precisely when $\gcd(a, e) = 1$. Choose a with the class of x^a equal to χ ; then $\mathcal{Q} \cong \iota_*(T^e = x^a)$ over $k((x))$. The same equation over $k[x, x^{-1}]$ defines a $\mathbb{Z}/e\mathbb{Z}$ -torsor over $\mathbb{G}_m$ (étale since e is invertible in k), so $\mathcal{P}' := \iota_*(\text{Spec}(k[x, x^{-1}][T]/(T^e - x^a)))$ is a G -torsor over $\mathbb{G}_m$ with $\mathcal{P}'_x \cong \mathcal{Q}$; it is tamely ramified at 0 and at ∞ since its local characters there factor through the tame quotients.

(2) If $\mathcal{Q}$ is trivial, take $f = 0$. Otherwise, since $\mathbb{Z}/p\mathbb{Z}$ is simple and $k((x))$ admits no non-trivial unramified extensions, $\mathcal{Q} = \text{Spec}(F)$ for a cyclic extension $F/k((x))$ of degree p together with an isomorphism $\text{Gal}(F/k((x))) \cong \mathbb{Z}/p\mathbb{Z}$. By Artin–Schreier theory such data correspond to non-zero classes in $k((x))/\wp(k((x)))$, where $\wp(u) = u^p - u$, via $g \mapsto k((x))[X]/(X^p - X - g)$. We claim that every class admits a representative f as in the statement:

- $\wp$ is surjective on $k[[x]]$: given $h = \sum_{i \geq 0} b_i x^i$, solve $\wp(u) = h$ with $u = \sum_{i \geq 0} u_i x^i$ by choosing u_0 with $u_0^p - u_0 = b_0$ (possible as k is algebraically closed) and setting recursively $u_i = u_{i/p}^p - b_i$, with the convention $u_{i/p} = 0$ when $p \nmid i$. Hence, we may choose a representative in $x^{-1}k[x^{-1}]$.
- If the representative has pole order m divisible by p , with leading term cx^{-m} , then subtracting $\wp(c^{1/p}x^{-m/p}) = cx^{-m} - c^{1/p}x^{-m/p}$ strictly decreases the pole order. Iterating, we reach a representative f which is either 0 (impossible, as the class is non-zero) or has pole order m prime to p .

By Theorem 14, the unique ramification jump of $F = k((x))[X]/(X^p - X - f)$ is at m , so the ramification of $\mathcal{Q}$ is bounded by d if and only if $m < d$. Finally, $f \in k[x, x^{-1}]$, so the same equation defines a finite étale $\mathbb{Z}/p\mathbb{Z}$ -cover $\mathcal{P}' \rightarrow \mathbb{G}_m$ (a torsor via $X \mapsto X + 1$) with $\mathcal{P}'_x \cong \mathcal{Q}$; and writing $y = x^{-1}$, we have $f \in yk[y]$, so $X^p - X - f$ defines a finite étale cover of $\mathbb{A}_k^1 = \text{Spec}(k[y]) = \mathbb{P}_k^1 \setminus \{0\}$ extending $\mathcal{P}'$, whence $\mathcal{P}'$ is unramified at ∞ . $\square$

Combining [Corollary 30](#) with [Lemma 31](#), in order to prove [Theorem 1](#) for a torsor which is tamely ramified at P , or for a $\mathbb{Z}/p\mathbb{Z}$ -torsor with ramification bounded by d at P , we may — and do — assume from now on that

$$C = \mathbb{P}_k^1, \quad \mathfrak{m} = d \cdot 0 + \infty, \quad \mathfrak{n} = d \cdot 0, \quad U = \mathbb{G}_m,$$

and that $\mathcal{P} \rightarrow \mathbb{G}_m$ is one of the explicit torsors of [Lemma 31](#).

1.3 The symmetric-power computation: $\mathbb{Z}/e\mathbb{Z}$, $\mathbb{Z}/p\mathbb{Z}$, $\mathbb{Z}/p^r\mathbb{Z}$

Having reduced to an explicit torsor over $\mathbb{G}_m$, the computation now proceeds identically in the three cases, along the following four steps.

Step A (realization). Present the local torsor $\mathcal{P}_x$ by an explicit equation over $\mathbb{G}_m$ — Kummer $T^e = x^a$, Artin–Schreier $X^p - X = f$, or Artin–Schreier–Witt $\wp \vec{X} \ominus \vec{f}$ — with the pole data of the defining datum controlled by the ramification bound.

Step B (invariant ring). On $C^{\times d}$, form the d -fold product torsor and identify the ring of K_G -invariants of $S^{\otimes kd}$: it is generated by a single invariant Y , namely $\prod_i X_i$ (Kummer), $\sum_i X_i$ (Artin–Schreier), or the Witt sum $\bigoplus_i \vec{X}_i$ (Artin–Schreier–Witt).

Step C (descent to e_i). Pass to S_d -invariants: the resulting defining datum is a symmetric function of $x_1, \dots, x_d$, hence a function of the elementary symmetric coordinates $e_1, \dots, e_d$.

Step D (regularity at η). Because the pole order of the datum is $< d$, it is regular at η_n ; the cover $\mathcal{P}^{[d]}|_{\hat{K}_\eta}$ is therefore tame (Kummer) or unramified (Artin–Schreier and Artin–Schreier–Witt).

The Kummer and Artin–Schreier cases are carried out together in [Theorem 32](#); the Artin–Schreier–Witt case, which occupies the rest of the section, follows the same four steps in Witt coordinates. [Table 1.2](#) collects the three parallel calculations.

	Kummer	Artin–Schreier	Artin–Schreier–Witt
A: equation over $\mathbb{G}_m$	$T^e = x^a$	$X^p - X = f$	$\wp \vec{X} \ominus \vec{f}$
B: invariant Y	$\prod_i X_i$	$\sum_i X_i$	$\bigoplus_i \vec{X}_i$
B: $(S^{\otimes d})^{K_G}$	$R^{\otimes d}[Y]/(Y^e - \prod_i f_i)$	$R^{\otimes d}[Y]/(Y^p - Y - \sum_i f(x_i))$	$R^{\otimes d}[\vec{Y}]/(\wp \vec{Y} \ominus \vec{\alpha})$
C: datum in e_i	$\prod_i f_i = \tilde{F}(e)$	$\sum_i f(x_i) \in k(\frac{e_i}{e_d})$	$\vec{\alpha} \in W_r(k(\frac{e_i}{e_d}))$
D: bound at η	pole $< d$	pole $m < d$	$p^{r-1-j}m_j < d$
D: conclusion	tame	unramified	unramified

Table 1.2: The three symmetric-power computations in parallel (Steps A–D).

We start by analyzing the blowup $\tilde{X} = \text{Bl}_0(X)$ of $X = \mathbb{A}_k^n$ where $0 \in X$ be the origin. Recall that $\tilde{X} \subset X \times_k \mathbb{P}_k^{n-1}$ is defined by the equations $x_i u_j = x_j u_i$, where $[u_1 : \dots : u_n]$ are the homogeneous coordinates of $\mathbb{P}_k^{n-1}$. The exceptional divisor $E \subset \tilde{X}$ is the fiber over the origin, $E = \{(0, [u_1 : \dots : u_n])\}$, which is of codimension 1 in $\tilde{X}$. Let $\eta \in E$ be the generic point of E , and let $R = \mathcal{O}_{\tilde{X}, \eta}$ be the associated local ring. This ring R is a discrete valuation ring (DVR) with fraction field $K = K(X) = k(x_1, \dots, x_n)$.

On the affine chart U_1 where $u_1 \neq 0$, we have $x_i = \frac{u_i}{u_1}x_1$. The coordinate ring is:

$$\mathcal{O}_{\hat{X}}(U_1) = k\left[x_1, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1}\right]$$

In this chart, the generic point η corresponds to the prime ideal $\mathfrak{p}_1 = (x_1)$. Thus, the local ring is $R = k[x_1, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1}]_{(x_1)}$. The residue field is $\kappa(\eta) = k(\frac{u_2}{u_1}, \dots, \frac{u_n}{u_1})$, and the completion of R with respect to its maximal ideal is:

$$\hat{R} = \kappa(\eta)[[x_1]]$$

In this local ring, x_1 is a uniformizer. Note that any x_i (for $i > 1$) can also serve as a uniformizer, as $x_i = (\frac{u_i}{u_1})x_1$ and $\frac{u_i}{u_1}$ is a unit in R .

For any monomial $M = x_1^{a_1} \dots x_n^{a_n} \in K$, we can write:

$$M = x_1^{\sum a_i} \left(\frac{u_2}{u_1}\right)^{a_2} \dots \left(\frac{u_n}{u_1}\right)^{a_n}$$

Since the term in parentheses is a unit in R , the valuation ν_E associated with E satisfies:

$$\nu_E(M) = \sum a_i = \deg M$$

Consequently, for any polynomial $f = f_d + f_{d+1} + \dots + f_l$, where f_i is the homogeneous part of degree i , we have $\nu_E(f) = d$ (the order of vanishing at the origin).

Our first result is:

Theorem 32. *Let $\mathcal{P} \rightarrow \mathbb{G}_m \subset \mathbb{P}_k^1$ be a G torsor which is either*

1. *tamely ramified at 0*
2. *totally ramified at 0 with $G = \mathbb{Z}/p\mathbb{Z}$ and ramification bounded by d .*

Then the ramification of the G -torsor $\mathcal{P}^{[d]} \rightarrow \mathbb{G}_m^{(d)}$ at η_m the generic point of $E_m \subset X_m$ is

1. *tamely ramified if $\mathcal{P}$ was tamely ramified*
2. *unramified if $\mathcal{P}$ was totally ramified with $G = \mathbb{Z}/p\mathbb{Z}$ and ramification bounded by d*

Proof. The local ring at the generic point of the exceptional divisor of the blowup of the affine space at 0 point is $R = k[x_d, \frac{u_1}{u_d}, \dots, \frac{u_{d-1}}{u_d}]_{(x_d)}$. Where for every $i < d$, we have $x_i = \frac{u_i}{u_d}x_d$ are all uniformizers. The residue field is $\kappa(\eta) = k(\frac{u_1}{u_d}, \dots, \frac{u_{d-1}}{u_d})$ and the completion of R with respect to its maximal ideal is:

$$\hat{R} = \kappa(\eta)[[x_d]]$$

In our situation, when we take symmetric product of the affine space, the situation is similar with different coordinates if we let $e_1, \dots, e_d$ be the symmetric polynomials in $x_1, \dots, x_d$ then: The local ring is $R = k[e_d, \frac{u_2}{u_d}, \dots, \frac{u_{d-1}}{u_d}]_{(e_d)}$. $e_i = \frac{u_i}{u_d}e_d$ are all uniformizers. The residue field being: $\kappa(\eta) = k(\frac{u_1}{u_d}, \dots, \frac{u_{d-1}}{u_d})$ and the completion: $\hat{R} = \kappa(\eta)[[s_d]]$ Note that from $e_i = \frac{u_i}{u_d}e_d$ We get $\frac{e_i}{e_d} = \frac{u_i}{u_d}$ in the fraction field. hence

$$\hat{K} = k\left(\frac{e_1}{e_d}, \dots, \frac{e_{d-1}}{e_d}\right)((e_d)) \quad (1.6)$$

We compute directly the extension of complete valued fields over the complete valued field at the generic point. Note that by [Corollary 30](#) and [Lemma 31](#) we can assume $\mathcal{P} = \text{Spec } k[x, x^{-1}][X]/(X^n - a)$ for $a \in k[x, x^{-1}]$ or $\mathcal{P} = \text{Spec } k[x, x^{-1}][X]/(X^p + X - f(x, x^{-1}))$ where $f(x, x^{-1}) = cx^{-m} + a_{-m+1}x^{-m+1} + \dots + a_{-1}x^{-1} + a_0 = cx^{-m} + f_{-m+1}(x^{-1})$ where $m < d$.

We deal with each case separately.

Artin-Schreier Extensions:

Set $R = k[x, x^{-1}]$, and $S = \text{Spec } k[x, x^{-1}][X]/(X^p + X - f(x^{-1}))$ we have

$$\begin{aligned} R^{\otimes_k d} &= k[x_1, x_1^{-1}, \dots, x_d, x_d^{-1}] \\ S^{\otimes_k d} &= k[x_1, x_1^{-1}, \dots, x_d, x_d^{-1}][X_1, \dots, X_d]/(X_1^p - X_1 - f(x_1), \dots, X_d^p - X_d - f(x_d)) \\ &= R^{\otimes_k d}[X_1, \dots, X_d]/(X_1^p - X_1 - f(x_1), \dots, X_d^p - X_d - f(x_d)) \end{aligned}$$

Next, we want to understand the ring corresponding to $p_1^{-1}(\mathcal{P}) \otimes \dots \otimes p_d^{-1}(\mathcal{P})$ on C^d - the d 'th-product of the $G = \mathbb{Z}/p\mathbb{Z}$ -torsors $p_1^{-1}(\mathcal{P}), \dots, p_d^{-1}(\mathcal{P})$ on U^d . It corresponds to quotient: $(p_1^{-1}(\mathcal{P}) \times \dots \times p_d^{-1}(\mathcal{P}))/G^{d-1}$ where the action of G^{d-1} on the product is:

$$(g_1, \dots, g_{d-1}) \cdot (p_1, p_2, \dots, p_{d-1}, p_d) = (g_1(p_1), g_1^{-1}g_2(p_2), \dots, g_{d-2}^{-1}g_{d-1}(p_{d-1}), g_{d-1}^{-1}(p_d))$$

The affine ring corresponding to the torsor product is $(S^{\otimes_k d})^{G^{d-1}}$

Recall that the action of $g \in G = \mathbb{Z}/p\mathbb{Z}$ on X is $g(X) = X + g$ (g correspond to a number $0 \leq g \leq p-1$). So, the action of $(g_1, \dots, g_{d-1})$ on the generators $(X_1, X_2, \dots, X_{d-1}, X_d)$ is $X_1 \mapsto X_1 + g_1$, $X_i \mapsto X_i - g_{i-1} + g_i$ for $1 < i < d$ and $X_d \mapsto X_d - g_{d-1}$. So we see that $Y = X_1 + \dots + X_d$ is invariant. Moreover $Y^p - Y - \sum_{i=1}^d f(x_i) = 0$ is irreducible degree p equation for Y , Since we are quotienting a rank p^d extension by a group of order p^{d-1} the resulting invariant subring must have rank p over $R^{\otimes_k d}$, So we conclude:

$$(S^{\otimes_k d})^{G^{d-1}} \cong R^{\otimes_k d}[Y]/(Y^p - Y - \sum_{i=1}^d f(x_i))$$

The group S_d acts on $R^{\otimes_k d} = k[x_1^{\pm 1}, x_2^{\pm 1}, \dots, x_d^{\pm 1}]$ by permuting the variables $\{x_i\}_{i=1}^d$. And since $Y = \sum_{i=1}^d X_i$ it leaves Y invariant. The invariant subring $(R^{\otimes_k d})^{S_d}$ is simply $k[e_1, e_2, \dots, e_d, e_d^{-1}]$ where $\{e_i\}$ are the symmetric polynomials in $x_1, \dots, x_d$:

$$e_k = \sum_{1 \leq j_1 < j_2 < \dots < j_k \leq d} x_{j_1} x_{j_2} \dots x_{j_k}$$

To find $(R^{\otimes_k d}[Y]/(Y^p - Y - \sum_{i=1}^d f(x_i)))^{S_d}$ its enough to express $\sum_{i=1}^d f(x_i)$ in $e_1, \dots, e_d$, this can be done with the newton polynomials, moreover, we claim the following:

Lemma 33. *Let $f(x) = cx^{-m} + a_{-m+1}x^{-m+1} + \dots + a_{-1}x^{-1} + a_0$, define $\alpha(x_1, \dots, x_d) = \sum_{i=1}^d f(x_i)$, and denote by $e_1, \dots, e_d$ the elementary symmetric polynomials in $x_1, \dots, x_d$. If $m < d$ then $\alpha(x_1, \dots, x_d) \in k(e_1/e_d, e_2/e_d, \dots, e_{d-1}/e_d)$*

Proof. Changing variables $y_i = x_i^{-1}$ for each $i \in \{1, \dots, d\}$ We get

$$\alpha = \sum_{i=1}^d f(x_i) = \sum_{i=1}^d \left(cy_i^m + a_{-m+1}y_i^{m-1} + \dots + a_{-1}y_i + a_0 \right)$$

Rearranging the sums, we get

$$\alpha = c \sum_{i=1}^d y_i^m + a_{-m+1} \sum_{i=1}^d y_i^{m-1} + \dots + a_{-1} \sum_{i=1}^d y_i + da_0$$

Let $p_k(y_1, \dots, y_d) = \sum_{i=1}^d y_i^k$ be the k -th power sum symmetric polynomial. The expression for α is a linear combination of these power sums:

$$\alpha = cp_m(y) + a_{-m+1}p_{m-1}(y) + \dots + a_{-1}p_1(y) + da_0$$

According to the *Fundamental Theorem of Symmetric Polynomials*, any symmetric polynomial in $y_1, \dots, y_d$ can be expressed as a polynomial in the elementary symmetric polynomials $e_k(y_1, \dots, y_d)$. Since $m < d$, α is a polynomial in $e_1(y), e_2(y), \dots, e_m(y)$. ($y = (y_1, \dots, y_d)$) The elementary symmetric polynomials in $y_i = 1/x_i$ are related to the elementary symmetric polynomials in x_i as follows:

$$e_k(y_1, \dots, y_d) = \sum_{1 \leq i_1 < \dots < i_k \leq d} \frac{1}{x_{i_1} \dots x_{i_k}} = \frac{\sum_{1 \leq j_1 < \dots < j_{d-k} \leq d} x_{j_1} \dots x_{j_{d-k}}}{x_1 x_2 \dots x_d}$$

Thus,

$$e_k(y_1, \dots, y_d) = \frac{e_{d-k}(x_1, \dots, x_d)}{e_d(x_1, \dots, x_d)}$$

which concludes the proof. $\square$

Finally, restricting $\mathcal{P}^{[d]}$ to $\text{spec } \hat{K}$ we get by (1.6) and Theorem 14 the result. (that $\mathcal{P}$ is unramified at the generic point of the exceptional divisor of the blowup).

Kummer Extensions: Few things are different in that case,

Set $R = k[x, x^{-1}]$, and $S = R[X]/(X^n - f)$ where $f = f(x, 1/x) \in R$ In this case we have $\text{char } k = p$ and $\gcd(p, n) = 1$.

We have

$$\begin{aligned} R^{\otimes kd} &= k[x_1, x_1^{-1}, \dots, x_d, x_d^{-1}] \\ S^{\otimes kd} &= k[x_1, x_1^{-1}, \dots, x_d, x_d^{-1}][X_1, \dots, X_d]/(X_1^n - f_1, \dots, X_d^n - f_d) \\ &= R^{\otimes kd}[X_1, \dots, X_d]/(X_1^n - f_1, \dots, X_d^n - f_d) \end{aligned}$$

Where $f_i = f(x_i, x_i^{-1})$

Next, we want to figure out $(S^{\otimes kd})^{G^{d-1}}$

Recall that the action of $g \in G = \mathbb{Z}/n\mathbb{Z}$ on X is $g(X) = \zeta^g X$ (g correspond to a number $0 \leq g \leq n-1$). So, the action of $(g_1, \dots, g_{d-1})$ on the generators $(X_1, X_2, \dots, X_{d-1}, X_d)$ Is $X_1 \mapsto \zeta^{g_1} X_1$, $X_i \mapsto \zeta^{g_i - g_{i-1}} X_i$ for $1 < i < d$ and $X_d \mapsto \zeta^{-g_{d-1}} X_d$.

So we see that $Y = X_1 X_2 \dots X_d$ is invariant. And $Y^n - \prod_{i=1}^d f_i$ is irreducible degree n equation for Y , So, like before, we conclude:

$$\left(S^{\otimes_k d}\right)^{G^{d-1}} \cong R^{\otimes_k d}[Y]/(Y^n - \prod_{i=1}^d f_i)$$

The group S_d acts on $R^{\otimes_k d}[Y]/(Y^n - \prod_{i=1}^d f_i)$ by permuting the indices. on the variables x_i , On $Y = \prod_{i=1}^d X_i$ it is invariant. The invariant subring $(R^{\otimes_k d})^{S_d}$ is simply $k[e_1, e_2, \dots, e_d, e_d^{-1}]$ like before.

The polynomial $F = \prod_{i=1}^d f_i$ is symmetric in $\{x_i\}_1^d$ so it can be expressed as a polynomial $\tilde{F}(e_1, \dots, e_d)$ in the elementary symmetric variables. Hence the quotient ring is:

$$\left(\frac{k[x_1^{\pm 1}, \dots, x_d^{\pm 1}][Y]}{(Y^n - \prod_{i=1}^d f_i)}\right)^{S_d} \cong \frac{k[e_1, \dots, e_d, e_d^{-1}][Y]}{(Y^n - \tilde{F}(e_1, \dots, e_d))}$$

So we see again, that restricting $\mathcal{P}^{[d]}$ to $\text{spec } \hat{K}$ we get by (1.6) and Theorem 13, that $\mathcal{P}$ is tamely ramified at the generic point of the exceptional divisor of the blowup. $\square$

So, we conclude

Corollary 34. *In the general settings of a curve C , and $\mathcal{P} \rightarrow U = C \setminus \mathfrak{m}$ a G -torsor with ramification bounded by $\mathfrak{n} = nP \subset \mathfrak{m}$ at P , we have:*

1. *If $G = \mathbb{Z}/p\mathbb{Z}$ then $\mathcal{P}^{[\deg \mathfrak{n}]}$ is unramified at $\eta_{\mathfrak{n}}$*
2. *If $\mathcal{P}$ is tamely ramified at $\mathfrak{n}$ then $\mathcal{P}^{[\deg \mathfrak{n}]}$ is tamely ramified at $\eta_{\mathfrak{n}}$*

Proof. By the remark opening this section we may assume k is algebraically closed, so that $P \in C(k)$ and $\deg \mathfrak{n} = n$. The restriction $\mathcal{P}_x$ of $\mathcal{P}$ to the punctured formal disc at P is tamely ramified, resp. a $\mathbb{Z}/p\mathbb{Z}$ -torsor with ramification bounded by n , so by Lemma 31 there is a torsor $\mathcal{P}' \rightarrow \mathbb{G}_m$ of the corresponding explicit form with $\mathcal{P}'_x \cong \mathcal{P}_x$. By Corollary 30, the ramification of $\mathcal{P}^{[\deg \mathfrak{n}]}$ at $\eta_{\mathfrak{n}}$ agrees with that of $\mathcal{P}'^{[\deg \mathfrak{n}]}$ at $\eta_{n \cdot 0}$, which was computed in Theorem 32. $\square$

Next, we generalize to the case $G = \mathbb{Z}/p^r\mathbb{Z}$. This is the third column of Table 1.2, and it follows the same Steps A–D as the Kummer and Artin–Schreier cases of Theorem 32: the paragraph “Realization over $\mathbb{G}_m$ for Witt vectors” below is Step A, “the symmetric-power cover at $\eta_{\mathfrak{n}}$ ” is Steps B–C, and “the degree count” together with the final proof is Step D — each carried out in Witt coordinates using the theory of Section 1.1.3.

Theorem 35. *Assume $G = \mathbb{Z}/p^r\mathbb{Z}$. Then $\mathcal{P}^{[\deg(\mathfrak{n})]}$ is unramified at $\eta_{\mathfrak{n}}$.*

The proof occupies the remainder of this subsection. In contrast with what the phrase “we generalize” may suggest, the argument is *not* an induction on r with Corollary 34(1) as its base case: it is a direct argument, treating all r at once, in which each of the three steps of the Artin–Schreier case of Theorem 32 — realization of the local torsor by an explicit equation over $\mathbb{G}_m$, identification of the symmetric-power cover at $\eta_{\mathfrak{n}}$, and the degree estimate — is replaced by its Witt-vector analogue, using the theory collected in Section 1.1.3. In particular the case $r = 1$ is *reproved* (as the special case of Witt vectors of length one) rather than used. For an explanation of why the seemingly natural induction on r fails — in short: the hypothesis bounds the ramification of $\mathcal{P}$ in *upper* numbering, while the break of the top degree- p layer $\mathcal{P} \rightarrow \mathcal{P}'$ is the largest *lower*-numbering break, which can be of size about $p^{r-1}d$ — see Chapter A.

Standing reduction

As in the proof of [Corollary 34](#), we may assume k is algebraically closed. We may further assume that $\mathcal{P} \rightarrow U$ is *totally ramified* at P with inertia equal to all of G : writing $R = \widehat{\mathcal{O}_{C,P}}$, $K = \text{Frac}(R)$, and letting $H = \rho(G^0) \subseteq G$ be the image of the inertia subgroup under the homomorphism $\rho: \text{Gal}(K^{\text{sep}}/K) \rightarrow G$ classifying $\mathcal{P}|_{\text{Spec } K}$, decompose $\mathcal{P} \rightarrow \mathcal{P}/H \rightarrow U$ as in [Proposition 15](#), where $\mathcal{P} \rightarrow \mathcal{P}/H$ is an H -torsor and $\mathcal{P}/H \rightarrow U$ a G/H -torsor. Then $\mathcal{P}/H$ is unramified at P , hence extends over P ; taking any point $Q \in \mathcal{P}/H$ over P yields $\widehat{\mathcal{O}_{\mathcal{P}/H,Q}} \cong \widehat{\mathcal{O}_{C,P}}$, and since k is algebraically closed the torsor $\mathcal{P}/H$ is locally trivial at Q . This replaces $(G, \mathcal{P} \rightarrow U)$ by $(H, \mathcal{P} \rightarrow \mathcal{P}/H)$ with $H \cong \mathbb{Z}/p^s\mathbb{Z}$ for some $s \leq r$; as the theorem is proved for all r simultaneously, we may rename and assume $H = G$.

After this reduction, the restriction $\mathcal{Q} := \mathcal{P}_x$ of $\mathcal{P}$ to the punctured formal disc $\text{Spec } k((x))$ at P (with x a uniformizer at P) is a *connected* $\mathbb{Z}/p^r\mathbb{Z}$ -torsor: $\mathcal{Q} = \text{Spec } L$ for a cyclic, totally ramified extension L/K of degree p^r , with ramification bounded by $d := \deg \mathfrak{n}$ ([Definition 7](#)); equivalently, the largest upper-numbering break u of L/K satisfies $u < d$ *strictly*.

Realization over $\mathbb{G}_m$ for Witt vectors

The first step generalizes [Lemma 31\(2\)](#) from $\mathbb{Z}/p\mathbb{Z}$ to $\mathbb{Z}/p^r\mathbb{Z}$: every local torsor class as above is realized by an explicit Artin–Schreier–Witt equation over $\mathbb{G}_m$ whose Witt datum is a vector of *polynomials in x^{-1}* with controlled pole orders. Recall from [Definition 27](#) the notion of a reduced vector; we add:

Definition 36. A vector $\vec{g} = (g_0, \dots, g_{r-1}) \in W_r(k((x)))$ is a **reduced polynomial vector** if it is reduced ([Definition 27](#)) and moreover every $g_j \in x^{-1}k[x^{-1}]$; then $m(g_j) = \deg_{x^{-1}} g_j$, and $m(g_j) = 0$ if and only if $g_j = 0$.

Lemma 37 (surjectivity of $\wp$ on integral Witt vectors). $\wp: W_r(k[[x]]) \rightarrow W_r(k[[x]])$ is surjective.

Proof. Induction on r . For $r = 1$ this is the first bullet in the proof of [Lemma 31\(2\)](#): given $h = \sum_{i \geq 0} b_i x^i$, solve $u^p - u = h$ coefficientwise, using that k is algebraically closed for the constant term. For $r > 1$, let $\vec{h} \in W_r(k[[x]])$. By induction choose $\vec{u}' \in W_{r-1}(k[[x]])$ with $\wp \vec{u}' = t(\vec{h})$, and lift it to $\tilde{u} := (\vec{u}', 0) \in W_r(k[[x]])$. By [Lemma 22\(i\)](#), $t(\wp \tilde{u} \ominus \vec{h}) = \wp \vec{u}' \ominus t(\vec{h}) = 0$, so $\wp \tilde{u} \ominus \vec{h} = V^{r-1}(g)$ for some $g \in k[[x]]$. Choose $w \in k[[x]]$ with $\wp w = -g$ (case $r = 1$ again). Then by [Lemma 22\(ii\)](#) and additivity of V^{r-1} ,

$$\wp(\tilde{u} \oplus V^{r-1}(w)) = \wp \tilde{u} \oplus V^{r-1}(\wp w) = (\vec{h} \oplus V^{r-1}(g)) \oplus V^{r-1}(-g) = \vec{h}. \quad \square$$

Proposition 38 (reduced polynomial representatives). *Every class in $W_r(k((x)))/\wp W_r(k((x)))$ has a reduced polynomial representative: for every $\vec{f} \in W_r(k((x)))$ there exists $\vec{u} \in W_r(k((x)))$ such that $\vec{g} := \vec{f} \ominus \wp \vec{u}$ has all components in $x^{-1}k[x^{-1}]$ with pole orders either 0 or prime to p .*

Proof. Induction on r .

Base case $r = 1$. This is the argument of [Lemma 31\(2\)](#): first, writing $f = f^- + f^+$ with $f^- \in x^{-1}k[x^{-1}]$ and $f^+ \in k[[x]]$, surjectivity of $\wp$ on $k[[x]]$ removes f^+ ; then, as long as the leading term is cx^{-m} with $p \mid m$, $m > 0$, subtracting $\wp(c^{1/p}x^{-m/p}) = cx^{-m} - c^{1/p}x^{-m/p}$ strictly decreases the pole order while staying inside $x^{-1}k[x^{-1}]$; this terminates at 0 or at pole order prime to p .

Inductive step. Let $\vec{f} \in W_r(k((x)))$. By induction there is $\vec{u}' \in W_{r-1}(k((x)))$ such that $\vec{g}' := t(\vec{f}) \ominus \wp \vec{u}'$ is a reduced polynomial vector in W_{r-1} . Lift $\vec{u}'$ to $\vec{u} := (\vec{u}', 0) \in W_r(k((x)))$ and set $\vec{h} := \vec{f} \ominus \wp \vec{u}$. By Lemma 22(i), $t(\vec{h}) = \vec{g}'$, i.e.

$$\vec{h} = (g'_0, \dots, g'_{r-2}, h_{r-1}), \quad h_{r-1} \in k((x)),$$

with the first $r-1$ components already reduced polynomials. It remains to repair the last component *without touching the others*; by Lemma 22(iii), adjustments of the form $\ominus \wp(V^{r-1}(z)) = \ominus V^{r-1}(\wp z)$ change only the last component, by $-\wp(z)$. So:

- Write $h_{r-1} = h^- + h^+$ with $h^- \in x^{-1}k[x^{-1}]$, $h^+ \in k[[x]]$, choose $w \in k[[x]]$ with $\wp w = h^+$ (Lemma 37 with $r = 1$), and replace $\vec{h}$ by $\vec{h} \ominus \wp(V^{r-1}w)$: the last component becomes h^- .
- While the last component has leading term cx^{-m} with $p \mid m$, $m > 0$, replace $\vec{h}$ by $\vec{h} \ominus \wp(V^{r-1}(c^{1/p}x^{-m/p}))$: the last component changes by $-\wp(c^{1/p}x^{-m/p}) = -cx^{-m} + c^{1/p}x^{-m/p}$, so its pole order strictly decreases and it stays in $x^{-1}k[x^{-1}]$. This terminates at 0 or at pole order prime to p .

The result is a reduced polynomial vector $\wp$ -equivalent to $\vec{f}$. □

Remark. The individual components of a Witt vector cannot be truncated to their principal parts one at a time — Witt addition is not componentwise ((1.3)), so a componentwise operation changes the Artin–Schreier–Witt class. The point of Proposition 38 is that all the adjustments are performed by subtracting genuine elements of $\wp W_r(k((x)))$, inside the Witt ring.

Lemma 39 (Realization over $\mathbb{G}_m$ for W_r). *Let k be algebraically closed of characteristic $p > 0$ and let $\mathcal{Q}$ be a connected $\mathbb{Z}/p^r\mathbb{Z}$ -torsor over $\text{Spec}(k((x)))$ with ramification bounded by d (Definition 7). Then there is a reduced polynomial vector $\vec{f} = (f_0, \dots, f_{r-1})$, $f_j \in x^{-1}k[x^{-1}]$, such that, with $m_j := \deg_{x^{-1}} f_j$:*

1. $p^{r-1-j} m_j < d$ for all $0 \leq j \leq r-1$; equivalently $m_j < dp^{j-(r-1)}$;
2. $\mathcal{P}' := \text{Spec}(k[x, x^{-1}][\vec{X}]/(\wp \vec{X} \ominus \vec{f}))$ is a $\mathbb{Z}/p^r\mathbb{Z}$ -torsor over $\mathbb{G}_m$ with $\mathcal{P}'_x \cong \mathcal{Q}$;
3. $\mathcal{P}'$ extends to a torsor over $\mathbb{P}_k^1 \setminus \{0\}$ — in particular it is unramified at ∞ — and has ramification bounded by d at 0.

Proof. $\mathcal{Q} = \text{Spec } L$ with L/K cyclic totally ramified of degree p^r and largest upper break $u < d$ (Theorem 24). Let $[\vec{f}_0] \in W_r(K)/\wp$ be its Artin–Schreier–Witt class and let $\vec{f}$ be a reduced polynomial representative (Proposition 38).

First, $f_0 \neq 0$: otherwise Corollary 25 would bound the order of the class by p^{r-1} , contradicting $[L : K] = p^r$. Hence $m_0 \geq 1$ and $p \nmid m_0$, and the Schmid–Witt break formula (Theorem 28) applies: $\max_j p^{r-1-j} m_j = u < d$, which is 1.

For 2: by Lemma 23 (with $A = k[x, x^{-1}] \ni f_j$), $\mathcal{P}'$ is a $\mathbb{Z}/p^r\mathbb{Z}$ -torsor over $\mathbb{G}_m$; its restriction to $\text{Spec } k((x))$ is the Artin–Schreier–Witt torsor of the class of $\vec{f}$, i.e. $\mathcal{Q}$ (Theorem 24).

For 3: in the coordinate $y = x^{-1}$ at ∞ we have $f_j \in y k[y]$, so $\vec{f} \in W_r(k[y])$ and Lemma 23 (with $A = k[y]$) extends $\mathcal{P}'$ to a torsor over $\mathbb{A}_k^1 = \text{Spec } k[y] = \mathbb{P}_k^1 \setminus \{0\}$; a torsor over the full local ring at ∞ is unramified there. The ramification bound at 0 holds because $\mathcal{P}'_x \cong \mathcal{Q}$. □

Remark. Connectedness of $\mathcal{Q}$ is guaranteed in the application by the standing reduction (inertia equal to all of G). Note also the following sanity check: for every j with $p^{r-1-j} \geq d$, the bound 1 forces $m_j = 0$, i.e. $f_j = 0$ — all sufficiently low Witt components of a reduced representative vanish outright. This makes visible how strong the hypothesis $u < d$ is at the lower levels.

The symmetric-power cover at η_n

We now generalize the invariant-ring computation in the Artin–Schreier case of [Theorem 32](#) to Witt coordinates. Fix $\vec{f} \in W_r(R)$, $R = k[x, x^{-1}]$, and let

$$S = R[\vec{X}]/(\wp \vec{X} \ominus \vec{f})$$

be the coordinate ring of the torsor $\mathcal{P}' \rightarrow \mathbb{G}_m$ of [Lemma 39](#). Write $R^{\otimes_k d} = k[x_1^{\pm 1}, \dots, x_d^{\pm 1}]$ and

$$S^{\otimes_k d} = R^{\otimes_k d}[\vec{X}_1, \dots, \vec{X}_d]/(\wp \vec{X}_1 \ominus \vec{f}(x_1), \dots, \wp \vec{X}_d \ominus \vec{f}(x_d)),$$

where $\vec{X}_i = (X_{i,0}, \dots, X_{i,r-1})$ and $\vec{f}(x_i)$ is the image of $\vec{f}$ under $x \mapsto x_i$. Exactly as in the proof of [Theorem 32](#), the d -fold product torsor $p_1^{-1}\mathcal{P}' \otimes^G \dots \otimes^G p_d^{-1}\mathcal{P}'$ on U^d is the quotient of $\text{Spec } S^{\otimes_k d}$ by $K_G \cong G^{d-1}$, embedded in G^d as the kernel of the sum map via $(g_1, \dots, g_{d-1}) \mapsto (g_1, g_2 - g_1, \dots, -g_{d-1})$, and acting (in Witt coordinates, writing $\underline{g}$ for the image of $g \in G$ under $\mathbb{Z}/p^r\mathbb{Z} \cong W_r(\mathbb{F}_p)$, [Example 20](#)) by

$$(g_1, \dots, g_{d-1}): \quad \vec{X}_1 \mapsto \vec{X}_1 \oplus \underline{g}_1, \quad \vec{X}_i \mapsto \vec{X}_i \oplus \underline{g}_i \ominus \underline{g}_{i-1} \quad (1 < i < d), \quad \vec{X}_d \mapsto \vec{X}_d \ominus \underline{g}_{d-1}.$$

Proposition 40 (identification of the product torsor). *Set*

$$\vec{Y} := \bigoplus_{i=1}^d \vec{X}_i \in W_r(S^{\otimes_k d}), \quad \vec{\alpha} := \bigoplus_{i=1}^d \vec{f}(x_i) \in W_r(R^{\otimes_k d}),$$

where $\bigoplus$ denotes the d -fold Witt vector sum $\oplus$ (not a direct sum); in particular $\vec{Y}$ is a single Witt vector of length r . Then:

1. Every component Y_n of $\vec{Y}$ is G^{d-1} -invariant, and

$$(S^{\otimes_k d})^{G^{d-1}} \cong R^{\otimes_k d}[\vec{Y}]/(\wp \vec{Y} \ominus \vec{\alpha})$$

as G -torsors over $R^{\otimes_k d}$ (the residual $G = G^d/G^{d-1}$ acting by $\vec{Y} \mapsto \vec{Y} \oplus \underline{g}$).

2. Every component α_n of $\vec{\alpha}$ is a symmetric Laurent polynomial, i.e. lies in $(R^{\otimes_k d})^{S_d} = k[e_1, \dots, e_d, e_d^{-1}]$, and

$$((S^{\otimes_k d})^{G^{d-1}})^{S_d} \cong k[e_1, \dots, e_d, e_d^{-1}][\vec{Y}]/(\wp \vec{Y} \ominus \vec{\alpha}).$$

Consequently, restricting to the complete local field $\widehat{K}_\eta = \kappa(\eta)((e_d))$, $\kappa(\eta) = k(e_1/e_d, \dots, e_{d-1}/e_d)$, of [\(1.6\)](#),

$$\mathcal{P}'^{[d]}|_{\text{Spec } \widehat{K}_\eta} \cong \text{Spec } \widehat{K}_\eta[\vec{Z}]/(\wp \vec{Z} \ominus \vec{\alpha}).$$

Proof. 1 An element $\sigma = (g_1, \dots, g_{d-1}) \in G^{d-1}$ acts as a ring automorphism of $S^{\otimes_k d}$: this is its *original* action, defined on the generators $X_{i,n}$ by the twisted formulas above (which mix components through the carry polynomials of (1.3)). By functoriality of W_r (Section 1.1.3), the induced map $W_r(\sigma)$ on $W_r(S^{\otimes_k d})$ is componentwise and commutes with $\oplus, \ominus$; on the generators it reproduces the twisted action, $W_r(\sigma)(\vec{X}_i) = \vec{X}_i \oplus \underline{g}_i \ominus \underline{g}_{i-1}$ (with $\underline{g}_0 = \underline{g}_d = \vec{0}$). Hence, using commutativity and associativity of $\oplus$, the Witt sum transforms by the telescoping constant

$$W_r(\sigma)(\vec{Y}) = \bigoplus_{i=1}^d W_r(\sigma)(\vec{X}_i) = \vec{Y} \oplus \underline{g}_1 \oplus (\underline{g}_2 \ominus \underline{g}_1) \oplus \dots \oplus (\ominus \underline{g}_{d-1}) = \vec{Y}.$$

Since $W_r(\sigma)$ is componentwise, this equality of r -tuples says exactly $\sigma(Y_n) = Y_n$ for every n , i.e. each $Y_n \in S^{\otimes_k d}$ is invariant. Since $\wp = F - \text{id}$ is additive (Definition 21 ff.),

$$\wp \vec{Y} = \bigoplus_{i=1}^d \wp \vec{X}_i = \bigoplus_{i=1}^d \vec{f}(x_i) = \vec{\alpha}.$$

Hence there is a well-defined $R^{\otimes_k d}$ -algebra homomorphism

$$\phi: T := R^{\otimes_k d}[\vec{Z}]/(\wp \vec{Z} \ominus \vec{\alpha}) \longrightarrow (S^{\otimes_k d})^{G^{d-1}}, \quad \vec{Z} \mapsto \vec{Y}.$$

By Lemma 23, $\text{Spec } T$ is a G -torsor over U^d . The target is the product torsor (the contracted product of the $p_i^{-1}\mathcal{P}'$ along the sum map $G^d \rightarrow G$), a G -torsor over U^d by Proposition 25 and the surrounding discussion; its residual G -action is induced by $(g, 0, \dots, 0) \in G^d$, i.e. $\vec{X}_1 \mapsto \vec{X}_1 \oplus \underline{g}$, under which $\vec{Y} \mapsto \vec{Y} \oplus \underline{g}$. So $\text{Spec } \phi$ is a G -equivariant morphism of G -torsors over U^d , hence an isomorphism by Proposition 5.

2 S_d acts on $S^{\otimes_k d}$ by permuting the tensor factors, i.e. $x_i \mapsto x_{\sigma(i)}$, $\vec{X}_i \mapsto \vec{X}_{\sigma(i)}$. As in 1, each $\sigma \in S_d$ is a ring automorphism, so $W_r(\sigma)$ is componentwise and commutes with $\oplus$. Since $\oplus$ is commutative and associative, permuting the summands of a Witt sum leaves it unchanged; thus $\vec{Y}$ is S_d -invariant (componentwise, as before) and $W_r(\sigma)(\vec{\alpha}) = \bigoplus_i \vec{f}(x_{\sigma(i)}) = \vec{\alpha}$, so each $\alpha_n \in R^{\otimes_k d}$ is a symmetric Laurent polynomial, hence lies in $k[e_1, \dots, e_d, e_d^{-1}]$ (the S_d -invariants of $R^{\otimes_k d}$, as in the proof of Theorem 32). By Lemma 23, T is free over $R^{\otimes_k d}$ with basis the monomials $\prod_n Y_n^{a_n}$, $0 \leq a_n < p$, each of which is S_d -invariant; writing an element in this basis, it is S_d -invariant if and only if all its coefficients are. Therefore

$$T^{S_d} = (R^{\otimes_k d})^{S_d}[\vec{Y}]/(\wp \vec{Y} \ominus \vec{\alpha}) = k[e_1, \dots, e_d, e_d^{-1}][\vec{Y}]/(\wp \vec{Y} \ominus \vec{\alpha}).$$

The final statement follows exactly as in the $r = 1$ case: by definition $\mathcal{P}'^{[d]}$ is the quotient by $\Xi = K_G \rtimes S_d$ computed above (Corollary 27), and its restriction to $\text{Spec } \hat{K}_\eta$ along $k[e_1, \dots, e_d, e_d^{-1}] \rightarrow \hat{K}_\eta$ is the Artin–Schreier–Witt cover of the image of $\vec{\alpha}$ in $W_r(\hat{K}_\eta)$. $\square$

Remark. The proof via Proposition 5 replaces the rank count of the $r = 1$ argument ($p^{rd}/p^{r(d-1)} = p^r$ together with irreducibility of the invariant’s minimal equation); the rank count alone would still require showing that $\vec{Y}$ generates the invariant ring, which is exactly what equivariance gives for free. Note also that a pure restriction argument on H^1 ’s (“the class of the product torsor is the sum of the pulled-back characters”) identifies the cover over $k(x_1, \dots, x_d)$ but does *not* by itself descend it to the symmetric base $k(e_1, \dots, e_d)$; the invariant-theoretic computation above is what effects the descent.

The degree count

Throughout this paragraph, $\vec{f}$ is the reduced polynomial vector of [Lemma 39](#), so $m_j = \deg_{x^{-1}} f_j$ satisfies $m_j < dp^{j-(r-1)}$, and $y_i := 1/x_i$, so that $f_j(x_i) \in y_i k[y_i]$ is a polynomial in y_i of degree m_j .

Lemma 41 (symmetric regularity). *Let $\Phi \in k[y_1, \dots, y_d]^{S_d}$ be symmetric with every monomial of total degree $< d$. Then $\Phi \in \kappa(\eta) = k(e_1/e_d, \dots, e_{d-1}/e_d)$; in particular Φ is regular at η_n .*

Proof. Φ is a k -combination of monomial symmetric functions $m_\lambda(y)$ with $|\lambda| < d$. Each $m_\lambda(y)$ is an isobaric polynomial in the elementary symmetric functions $e_\mu(y) = \prod_i e_{\mu_i}(y)$ with $\mu \vdash |\lambda|$; every part $\mu_i \leq |\lambda| < d$, so only $e_1(y), \dots, e_{d-1}(y)$ occur — never $e_d(y)$. Now $e_k(y) = e_{d-k}(x)/e_d(x)$, which for $1 \leq k \leq d-1$ lies in $\kappa(\eta)$. Hence $\Phi \in \kappa(\eta) \subset \hat{\mathcal{O}}_\eta = \kappa(\eta)[[e_d]]$. $\square$

Theorem 42 (regularity of all Witt components). *With $\vec{f}$ as in [Lemma 39](#), every component α_n ($0 \leq n \leq r-1$) of $\vec{\alpha} = \bigoplus_{i=1}^d \vec{f}(x_i)$ lies in $\kappa(\eta)$; in particular $\vec{\alpha} \in W_r(\hat{\mathcal{O}}_\eta)$.*

Proof. Fix $n \leq r-1$ and a monomial $\prod_{i,j} (f_j(x_i))^{e_{ij}}$ of α_n , viewed as a universal isobaric polynomial in the $f_j(x_i)$ of weight p^n ([Lemma 26](#), with $f_j(x_i)$ of weight p^j); set $E_j := \sum_i e_{ij}$, so $\sum_j E_j p^j = p^n$. Expanding each $f_j(x_i)$ as a polynomial in y_i of degree m_j , every resulting monomial in the y_i has total degree

$$\leq \sum_{i,j} e_{ij} m_j = \sum_j E_j m_j < \sum_j E_j dp^{j-(r-1)} = dp^{-(r-1)} \sum_j E_j p^j = dp^{n-(r-1)} \leq d,$$

strictly: some $E_j > 0$ (as $\sum_j E_j p^j = p^n > 0$), and the pole bound $m_j < dp^{j-(r-1)}$ of [Lemma 39](#) is strict even when $m_j = 0$. Thus every monomial of α_n — a symmetric polynomial in $y_1, \dots, y_d$ by [Proposition 402](#) — has total y -degree $< d$, and [Lemma 41](#) gives $\alpha_n \in \kappa(\eta)$. $\square$

Remark. The bound is exact: the weight $p^{j-(r-1)}$ is paired against the isobaricity constraint $\sum_j E_j p^j = p^n$, so the estimate $dp^{n-(r-1)} \leq d$ is independent of how the Witt carries distribute the exponents E_j ; the hypothesis “break $< d$ at P ” is used at *every* Witt level at once. The sanity check of [Lemma 39](#) records the extreme case: components with $p^{r-1-j} \geq d$ vanish outright in the reduced representative.

Proof of the theorem

Proof of Theorem 35. By the standing reduction above we may assume k algebraically closed and $\mathcal{P} \rightarrow U$ totally ramified at P with inertia all of G , so that $\mathcal{Q} := \mathcal{P}_x$ is a connected $\mathbb{Z}/p^r\mathbb{Z}$ -torsor over $\text{Spec } k((x))$ with ramification bounded by $d = \deg n$.

[Lemma 39](#) produces a torsor $\mathcal{P}' \rightarrow \mathbb{G}_m$, defined by a reduced polynomial vector $\vec{f}$ with $p^{r-1-j}m_j < d$, with $\mathcal{P}'_x \cong \mathcal{Q}$. By [Corollary 30](#) (valid for arbitrary finite abelian G), the ramification of $\mathcal{P}^{[d]}$ at η_n agrees with that of $\mathcal{P}'^{[d]}$ at $\eta_{d \cdot 0}$; so we may assume $C = \mathbb{P}_k^1$, $\mathfrak{m} = d \cdot 0 + \infty$, $\mathfrak{n} = d \cdot 0$, $\mathcal{P} = \mathcal{P}'$.

By [Proposition 40](#),

$$\mathcal{P}'^{[d]}|_{\text{Spec } \hat{K}_\eta} \cong \text{Spec } \hat{K}_\eta[\vec{Z}]/(\wp \vec{Z} \ominus \vec{\alpha}), \quad \vec{\alpha} = \bigoplus_{i=1}^d \vec{f}(x_i),$$

with $\hat{K}_\eta = \kappa(\eta)((e_d))$ the complete local field at η_n ((1.6)) and $\hat{\mathcal{O}}_\eta = \kappa(\eta)[[e_d]]$ its valuation ring.

By [Theorem 42](#), $\vec{\alpha} \in W_r(\kappa(\eta)) \subset W_r(\widehat{\mathcal{O}}_\eta)$. Hence, by [Lemma 23](#) applied over $A = \widehat{\mathcal{O}}_\eta$, the Artin–Schreier–Witt cover $\wp \vec{Z} = \vec{\alpha}$ is finite étale over $\widehat{\mathcal{O}}_\eta$, and $\mathcal{P}^{[d]}|_{\text{Spec } \widehat{K}_\eta}$ is its generic fibre: every field factor of $\widehat{K}_\eta[\vec{Z}]/(\wp \vec{Z} \ominus \vec{\alpha})$ is the fraction field of a finite étale $\widehat{\mathcal{O}}_\eta$ -algebra, i.e. an unramified extension of $\widehat{K}_\eta$ with respect to the discrete valuation ring $\widehat{\mathcal{O}}_\eta$. By [Definition 6](#), $\mathcal{P}^{[d]}$ is unramified at η_n . $\square$

Remark. The argument proves all r simultaneously; it reproves the case $r = 1$ ([Corollary 34\(1\)](#)) as the special case of a vector of length one and never invokes it.

Finally, we assemble the general finite abelian case from the three cyclic computations above.

Proof of Theorem 1. By the structure theorem, $G \cong \prod_i G_i$ with each G_i cyclic of prime-power order $\ell_i^{a_i}$; write $\pi_i : G \rightarrow G_i$ for the projections. The π_i jointly embed $G \hookrightarrow \prod_i G_i$, so the character $\rho : \text{Gal}(\widehat{K}_{\eta_n}^{\text{sep}}/\widehat{K}_{\eta_n}) \rightarrow G$ classifying $\mathcal{P}^{[d]}$ at η_n is tamely ramified as soon as each component $\pi_i \circ \rho$ is: a homomorphism into $\prod_i G_i$ kills the wild inertia iff each of its projections does. Now $\pi_i \circ \rho$ classifies the pushforward $\pi_{i,*}(\mathcal{P}^{[d]})$, and the contracted product along π_i commutes with the symmetric-power construction — both are built from fibre products and finite quotients ([Corollary 27](#), [Proposition 9](#)) — so

$$\pi_{i,*}(\mathcal{P}^{[d]}) \cong (\pi_{i,*}\mathcal{P})^{[d]}.$$

Here $\pi_{i,*}\mathcal{P}$ is a G_i -torsor whose character is $\pi_i \circ \rho$, so its ramification is again bounded by $\mathfrak{m}$ (composing with π_i can only shrink the image of the ramification groups). Applying the cyclic cases:

- if $\ell_i \neq p$, then $\pi_{i,*}\mathcal{P}$ is tamely ramified and $(\pi_{i,*}\mathcal{P})^{[d]}$ is tamely ramified at η_n by [Corollary 34](#);
- if $\ell_i = p$, then $G_i \cong \mathbb{Z}/p^{a_i}\mathbb{Z}$ and $(\pi_{i,*}\mathcal{P})^{[d]}$ is *unramified* at η_n by [Theorem 35](#).

In either case $\pi_{i,*}(\mathcal{P}^{[d]})$ is tamely ramified at η_n ; hence so is $\mathcal{P}^{[d]}$. $\square$

1.4 Ramification on Products of Blowups

We conclude this section by proving some auxiliary results that would be useful in [Chapter 3](#).

Let X and Y be smooth schemes over a field k , and let $x \in X$ and $y \in Y$ be closed points. We denote the blowups of these schemes at the given points by $\pi_X : \text{Bl}_x(X) \rightarrow X$ and $\pi_Y : \text{Bl}_y(Y) \rightarrow Y$. Furthermore, let $\pi_{X \times Y} : \text{Bl}_{(x,y)}(X \times_k Y) \rightarrow X \times_k Y$ be the blowup of the product scheme at the point (x, y) . We denote by E_X, E_Y , and $E_{X \times Y}$ the respective exceptional divisors, and let η_X, η_Y , and $\eta_{X \times Y}$ be their generic points.

In this section, we establish the following result concerning the stability of ramification bounds under the external product of torsors.

Proposition 43. *Let G be a finite abelian group. Suppose $\mathcal{G}_X$ and $\mathcal{G}_Y$ are G -torsors defined on open subsets $U_X \subset \text{Bl}_x(X)$ and $U_Y \subset \text{Bl}_y(Y)$ that are disjoint from the exceptional divisors. If the ramification of $\mathcal{G}_X$ at η_X and $\mathcal{G}_Y$ at η_Y is bounded by r , then the external product torsors*

$$\mathcal{G}_{X \times Y} := pr_1^{-1}\mathcal{G}_X \otimes pr_2^{-1}\mathcal{G}_Y$$

has ramification bounded by r at the generic point $\eta_{X \times Y}$ of the exceptional divisor in the product blowup.

The proposition is purely local in nature, it suffices to consider the case where X and Y are affine. More precisely, by the smoothness of X and Y , we may restrict our attention to open neighborhoods of x and y that are etale over affine spaces. The rest of this section treats that case.

The Affine Case

Let $X = \mathbb{A}_k^n$ be the affine n -space over a field k , and let $0 \in X$ be the origin. Let $\tilde{X} = \text{Bl}_0(X)$ be the blowup of X at the origin. Recall that $\tilde{X} \subset X \times_k \mathbb{P}_k^{n-1}$ is defined by the equations $x_i u_j = x_j u_i$, where $[u_1 : \cdots : u_n]$ are the homogeneous coordinates of $\mathbb{P}_k^{n-1}$. The exceptional divisor $E \subset \tilde{X}$ is the fiber over the origin, $E = \{(0, [u_1 : \cdots : u_n])\}$, which is of codimension 1 in $\tilde{X}$. Let $\eta \in E$ be the generic point of E , and let $R = \mathcal{O}_{\tilde{X}, \eta}$ be the associated local ring. This ring R is a discrete valuation ring (DVR) with fraction field $K = K(X) = k(x_1, \dots, x_n)$.

On the affine chart U_1 where $u_1 \neq 0$, we have $x_i = \frac{u_i}{u_1} x_1$. The coordinate ring is:

$$\mathcal{O}_{\tilde{X}}(U_1) = k \left[x_1, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1} \right]$$

In this chart, the generic point η corresponds to the prime ideal $\mathfrak{p}_1 = (x_1)$. Thus, the local ring is $R = k[x_1, \frac{u_2}{u_1}, \dots, \frac{u_n}{u_1}]_{(x_1)}$. The residue field is $\kappa(\eta) = k(\frac{u_2}{u_1}, \dots, \frac{u_n}{u_1})$, and the completion of R with respect to its maximal ideal is:

$$\hat{R} = \kappa(\eta)[[x_1]]$$

In this local ring, x_1 is a uniformizer. Note that any x_i (for $i > 1$) can also serve as a uniformizer, as $x_i = (\frac{u_i}{u_1})x_1$ and $\frac{u_i}{u_1}$ is a unit in R .

For any monomial $M = x_1^{a_1} \cdots x_n^{a_n} \in K$, we can write:

$$M = x_1^{\sum a_i} \left(\frac{u_2}{u_1} \right)^{a_2} \cdots \left(\frac{u_n}{u_1} \right)^{a_n}$$

Since the term in parentheses is a unit in R , the valuation ν_E associated with E satisfies:

$$\nu_E(M) = \sum a_i = \deg M$$

Consequently, for any polynomial $f = f_d + f_{d+1} + \cdots + f_l$, where f_i is the homogeneous part of degree i , we have $\nu_E(f) = d$ (the order of vanishing at the origin).

The Product Case

Now, let $X = \mathbb{A}^n$ and $Y = \mathbb{A}^m$ with origins $x = 0$ and $y = 0$. As before, $\text{Bl}_0(X) \subset X \times \mathbb{P}^{n-1}$ and $\text{Bl}_0(Y) \subset Y \times \mathbb{P}^{m-1}$ have exceptional divisors E_X and E_Y respectively. Consider the product $X \times_k Y \cong \mathbb{A}_k^{n+m}$. The blowup of the product at the origin $(0, 0)$, denoted $\text{Bl}_{(0,0)}(X \times_k Y)$, is a subscheme of $(X \times Y) \times \mathbb{P}^{n+m-1}$ defined by:

$$\begin{cases} x_i w_j = x_j w_i & 1 \leq i, j \leq n \\ y_k w_{n+l} = y_l w_{n+k} & 1 \leq k, l \leq m \\ x_i w_{n+j} = y_j w_i & 1 \leq i \leq n, 1 \leq j \leq m \end{cases}$$

where $[w_1 : \cdots : w_{n+m}]$ are the homogeneous coordinates of $\mathbb{P}^{n+m-1}$. The exceptional divisor $E_{X \times Y}$ is isomorphic to $\mathbb{P}^{n+m-1}$.

Comparison of Blowups

Both $\mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y)$ and $\mathrm{Bl}_{(0,0)}(X \times_k Y)$ are birational to $X \times_k Y$. They share a common dense open set $\tilde{U}$ defined by the condition that neither the X -coordinates nor the Y -coordinates vanish simultaneously in the projective space:

$$\tilde{U} = \{((x, y), [w_1 : \cdots : w_{n+m}]) \mid (w_1, \dots, w_n) \neq 0 \text{ and } (w_{n+1}, \dots, w_{n+m}) \neq 0\}$$

This yields a diagram of open immersions:

$$\begin{array}{ccc} & \tilde{U} & \\ f_1 \swarrow & & \searrow f_2 \\ \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y) & & \mathrm{Bl}_{(0,0)}(X \times_k Y) \end{array}$$

where f_1 maps the coordinates to the respective projectivizations $[w_1 : \cdots : w_n]$ and $[w_{n+1} : \cdots : w_{n+m}]$. Also note that the generic point $\eta_{X \times Y}$ of $E_{X \times Y}$ is inside $\tilde{U}$.

Extensions of DVRs

Let S be the local ring of the generic point $\eta_{X \times Y}$ of $E_{X \times Y}$ in $\tilde{U}$. On the chart where $w_1 \neq 0$ and $w_{n+1} \neq 0$, we have $x_1 = (\frac{w_1}{w_{n+1}})y_1$. Since $\frac{w_1}{w_{n+1}}$ is a unit in this chart, x_1 and y_1 are equivalent as uniformizers. We have:

$$S = k \left[x_1, \frac{w_2}{w_1} \cdots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1} \cdots \frac{w_{n+m}}{w_1} \right]_{(x_1)} = k \left[\frac{w_1}{w_{n+1}}, \frac{w_2}{w_{n+1}} \cdots \frac{w_n}{w_{n+1}}, y_1 \cdots \frac{w_{n+m}}{w_{n+1}} \right]_{(y_1)}$$

$$k(\eta_{X \times Y}) = k \left(\frac{w_2}{w_1} \cdots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1} \cdots \frac{w_{n+m}}{w_1} \right)$$

Let R_X be the local ring of the exceptional divisor E_X in $\mathrm{Bl}_0(X)$. The pullback of $E_X \times_k \mathrm{Bl}_0(Y)$ along f_1 induces an extension of DVRs $R_X \hookrightarrow S$. Which is:

1. Weakly Unramified: x_1 is a uniformizer in both R_X and S , so the ramification index is $e = 1$.
2. Residually Transcendental: The residue field extension $\kappa(\eta_X) \subset \kappa(\eta)$ is:

$$k \left(\frac{w_2}{w_1}, \dots, \frac{w_n}{w_1} \right) \subset k \left(\frac{w_2}{w_1}, \dots, \frac{w_n}{w_1}, \frac{w_{n+1}}{w_1}, \dots, \frac{w_{n+m}}{w_1} \right)$$

Hence separable.

Since this extension is generated by transcendental elements, it is separable and formally smooth at the maximal ideal ([Stacks, Tag 09E7]).

Ramification of G -Torsors

Let G be a finite abelian group. Let $\mathcal{Q}$ be a G -torsor on an open $U \subset \mathrm{Bl}_0(X)$ disjoint from E_X . Let $V \subset \mathrm{Bl}_0(Y)$ be an open subscheme, and let $\pi_X : \mathrm{Bl}_0(X) \times_k V \rightarrow \mathrm{Bl}_0(X)$ be the projection onto the first factor. By restricting this projection to $U \times_k V$, we obtain the pullback G -torsor:

$$\pi_X^{-1}(\mathcal{Q}) \cong \mathcal{Q} \times_k V$$

which is defined on the open subset $U \times_k V \subset \mathrm{Bl}_0(X) \times_k \mathrm{Bl}_0(Y)$. The extension of local rings $R_X \rightarrow S$ is weakly unramified (the ramification index $e = 1$) and residually transcendental with separable residue field extension. Under these conditions the ramification filtration is preserved. Therefore, the pullback $\pi_X^{-1}(\mathcal{Q})$ has ramification bounded by r at the generic point $\eta_{X \times Y}$ of the exceptional divisor $E_{X \times Y}$ if and only if the original torsor $\mathcal{Q}$ has ramification bounded by r at the generic point η_X of E_X .

And we finish by [Lemma 10](#).

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